INTERNET PROGRAMMING
LAB EXPERIMENTS

CSA4301

SLOT-D

UNIT-1

LAB PROGRAMS (1-6)

Name: Kala Gowri Shankar

Reg.No: 192325109

Experiment 1: Library Management System

Aim

To design a library management website with a book catalog and search, book details, user login and registration, borrow history with issue and due dates, fine calculation, study material by year and semester, and an admin dashboard to add, update and delete books, issue and return books, make fines and manage material.

HTML Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Library Management System</title>
<link
    href="https://fonts.googleapis.com/css2?family=Literata:wght@500;700&family=Public+Sans:wght@400;500;600&display=swap"
    rel="stylesheet">
<style>
:root{--bg:#f2f5f4;--surf:#fff;--ink:#14231f;--mute:#5b6b66;--line:#d5dfdc;--acc:#0f6b57;--acc-ink:#fff;--warn:#b42318;--soft:#e1ef
    eb;box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px)}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#0f1715;--surf:#18231f;--ink:#e7f0ed;--mute:#97aaa4;--li
    ne:#2b3b36;--acc:#4cc2a4;--acc-ink:#07201a;--warn:#ff8a7a;--soft:#1f3029}}
:root[data-theme="dark"]{--bg:#0f1715;--surf:#18231f;--ink:#e7f0ed;--mute:#97aaa4;--line:#2b3b36;--acc:#4cc2a4;--acc-ink:#0720
    1a;--warn:#ff8a7a;--soft:#1f3029}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--ink);font:15px/1.5 "Public Sans",system-ui,sans-serif}
h1,h2,h3{font-family:Literata,Georgia,serif;margin:0 0 .4em;line-height:1.2}
h1{font-size:clamp(28px,5vw,44px)}h2{font-size:24px}h3{font-size:18px}
header{background:var(--surf);border-bottom:1px solid var(--line);position:sticky;top:env(safe-area-inset-top,0px);z-index:5}
.bar{max-width:1080px;margin:auto;padding:12px 16px;display:flex;gap:8px;align-items:center;flex-wrap:wrap}
.brand{font:700 20px Literata,serif;margin-right:auto;cursor:pointer}
button,.btn{font:inherit;background:var(--acc);color:var(--acc-ink);border:0;border-radius:6px;padding:8px
    14px;cursor:pointer;font-weight:500}
button.ghost{background:transparent;color:var(--ink);border:1px solid var(--line)}
button.sm{padding:4px 10px;font-size:13px}button.del{background:var(--warn);color:#fff}
button:focus-visible,input:focus-visible,select:focus-visible,textarea:focus-visible{outline:2px solid var(--acc);outline-offset:2px}
main{max-width:1080px;margin:auto;padding:24px 16px 64px}
.hero{padding:28px 0 8px}.hero p{color:var(--mute);max-width:58ch}
input,select,textarea{font:inherit;width:100%;padding:9px 10px;border:1px solid
    var(--line);border-radius:6px;background:var(--surf);color:var(--ink)}
label{display:block;font-size:13px;color:var(--mute);margin:10px 0 4px}
.row{display:flex;gap:10px;flex-wrap:wrap}.row>*{flex:1;min-width:140px}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(230px,1fr));gap:14px;margin-top:16px}
.book{background:var(--surf);border:1px solid var(--line);border-left:6px solid
    var(--acc);border-radius:6px;padding:14px;cursor:pointer;display:flex;flex-direction:column;gap:4px}
.book:hover{border-color:var(--acc)}.book small{color:var(--mute)}
.tag{display:inline-block;font-size:12px;padding:2px 8px;border-radius:99px;background:var(--soft);color:var(--ink)}
.tag.no{background:transparent;border:1px solid var(--warn);color:var(--warn)}
.panel{background:var(--surf);border:1px solid var(--line);border-radius:8px;padding:18px;margin-top:14px}
.narrow{max-width:420px;margin:24px auto}
.tabs{display:flex;gap:6px;flex-wrap:wrap;margin:14px 0}
.tabs button{background:transparent;color:var(--ink);border:1px solid var(--line)}
.tabs button.on{background:var(--acc);color:var(--acc-ink);border-color:var(--acc)}
.tw{overflow-x:auto}table{border-collapse:collapse;width:100%;min-width:560px}
th,td{text-align:left;padding:8px;border-bottom:1px solid
    var(--line);vertical-align:top}th{font-size:13px;color:var(--mute);font-weight:600}
.late{color:var(--warn);font-weight:600}
```

```
.stats{display:flex;gap:12px;flex-wrap:wrap}.stat{flex:1;min-width:130px;background:var(--surf);border:1px solid var(--line);border-radius:8px;padding:12px}
.stat b{display:block;font:700 26px Literata,serif}
.modal{position:fixed;inset:0;background:rgba(0,0,0,.5);display:flex;align-items:center;justify-content:center;padding:16px;z-index:20}
.modal>div{background:var(--surf);border-radius:10px;padding:22px;max-width:520px;width:100%;max-height:85vh;overflow:auto}
#toast{position:fixed;left:50%;transform:translateX(-50%);bottom:calc(20px + env(safe-area-inset-bottom,0px));background:var(--ink);color:var(--bg);padding:10px 16px;border-radius:6px;z-index:30;display:none}
.muted{color:var(--mute)}
.i{width:1.05em;height:1.05em;vertical-align:-.17em;stroke:currentColor;fill:none;stroke-width:2;stroke-linecap:round;stroke-linejoin:round;margin-right:6px;flex:none}
.brand .i{color:var(--acc);width:22px;height:22px}.stat .i{color:var(--acc);width:20px;height:20px;margin:0 0 4px;display:block}
button .i{margin-right:5px}
</style></head><body>
<header><div class="bar" id="nav"></div></header>
<main id="app"></main>
<footer style="text-align:center;padding:14px;color:var(--mute);font-size:13px">Experiment 1: Library Management System | Name: Kala Gowri Shankar | Reg. No: 192325109</footer><div id="toast" role="status"></div><div id="mod"></div>
<script>
const K="gf_library_v1",RATE=5,LOAN=14;
const d0=()=>{const d=new Date();return d.getFullYear()+"-"+String(d.getMonth()+1).padStart(2,"0")+"-"+String(d.getDate()).padStart(2,"0")};
const addD=(s,n)=>{const d=new Date(s+"T00:00:00");d.setDate(d.getDate()+n);return d.getFullYear()+"-"+String(d.getMonth()+1).padStart(2,"0")+"-"+String(d.getDate()).padStart(2,"0")};
const diff=(a,b)=>Math.round((new Date(b+"T00:00:00")-new Date(a+"T00:00:00"))/864e5);
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
const $=id=>document.getElementById(id), v=id=>$(id).value.trim();
function seed(){const t=d0();return{nid:100,
users:[{id:1,name:"Mrs. Anita Rao",email:"admin@greenfield.edu",pw:"admin123",role:"admin"},
{id:2,name:"Priya Sharma",email:"priya@greenfield.edu",pw:"student123",role:"student",roll:"CS21045",dept:"Computer Science",year:2,sem:3},
{id:3,name:"Arjun Mehta",email:"arjun@greenfield.edu",pw:"student123",role:"student",roll:"ME21017",dept:"Mechanical",year:3,sem:5}],
books:[[1,"Introduction to Algorithms","T. H. Cormen","9780262046305","Computer Science",3,"The standard reference on algorithms, covering sorting, graphs, dynamic programming and more."],
[2,"Operating System Concepts","A. Silberschatz","9781119800361","Computer Science",2,"Processes, memory, storage and security in modern operating systems."],
[3,"Database System Concepts","A. Silberschatz","9780078022159","Computer Science",2,"Relational model, SQL, transactions and database design."],
[4,"Engineering Thermodynamics","P. K. Nag","9789339221324","Mechanical",2,"Laws of thermodynamics with worked engineering examples."],
[5,"Fundamentals of Fluid Mechanics","B. Munson","9781119080701","Mechanical",1,"Fluid statics, flow analysis and dimensional analysis."],
[6,"Higher Engineering Mathematics","B. S. Grewal","9788174091352","Mathematics",4,"Calculus, linear algebra, differential equations and numerical methods."],
[7,"Discrete Mathematics","K. H. Rosen","9780073383095","Mathematics",2,"Logic, sets, combinatorics, graphs and proof techniques."],
[8,"The God of Small Things","Arundhati Roy","9780812979657","Literature",1,"A novel set in Kerala about a family and the choices that shape it."]].map(a=>({id:a[0],title:a[1],author:a[2],isbn:a[3],cat:a[4],total:a[5],desc:a[6]})),
loans:[{id:1,bookId:2,userId:2,issue:addD(t,-25),due:addD(t,-11),ret:null,extra:0},
{id:2,bookId:1,userId:2,issue:addD(t,-40),due:addD(t,-26),ret:addD(t,-28),extra:0},
{id:3,bookId:4,userId:3,issue:addD(t,-5),due:addD(t,9),ret:null,extra:0}],
mats:[{id:1,title:"Data Structures: Unit 1 Notes",year:2,sem:3,body:"Arrays, linked lists, stacks and queues. Complexity analysis basics."},
{id:2,title:"DBMS Previous Year Questions",year:2,sem:3,body:"Normalization, SQL queries and transaction questions from the last five years."},
{id:3,title:"Thermodynamics Formula Sheet",year:3,sem:5,body:"First and second law relations, entropy and cycle efficiencies."},
{id:4,title:"Engineering Maths II Syllabus",year:1,sem:2,body:"Syllabus and reading list for semester 2."}],
msgs:[]}}
let db;try{const s=localStorage.getItem(K);db=s?JSON.parse(s):seed()}catch(e){db=seed()}
```

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const save=()=>{try{localStorage.setItem(K,JSON.stringify(db))}catch(e){}};
let me=null,view="home",tab="",q="",cat="",mq={y:"",s:"",t:""},edit=null;
const book=id=>db.books.find(b=>b.id==id),user=id=>db.users.find(u=>u.id==id);
const avail=b=>b.total-db.loans.filter(l=>l.bookId==b.id&&!l.ret).length;
const fine=l=>{const end=l.ret||d0();return Math.max(0,diff(l.due,end))*RATE+(l.extra||0)};
const IC={book:'<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/>',mail:'<rect x="2" y="4" width="20" height="16" rx="2"/><path d="m22 7-10 6L2 7"/>',user:'<circle cx="12" cy="8" r="4"/><path d="M4 21a8 8 0 0 1 16 0"/>',users:'<circle cx="9" cy="8" r="4"/><path d="M2 21a7 7 0 0 1 14 0"/><path d="M17 4a4 4 0 0 1 0 8M22 21a7 7 0 0 0 0-4-6"/>',login:'<path d="M15 3h4a2 2 0 0 1 2 2v14a2 2 0 0 1-2 2h-4"/><path d="m10 17 5-5-5-5"/><path d="M15 12H3"/>',logout:'<path d="M9 21H5a2 2 0 0 1-2-2V5a2 2 0 0 1 2-2h4"/><path d="m16 17 5-5-5-5"/><path d="M21 12H9"/>',plus:'<path d="M12 5v14M5 12h14"/>',edit:'<path d="M12 20h9"/><path d="M16.5 3.5a2.1 2.1 0 0 1 3 3L7 19l-4 1 1-4z"/>',trash:'<path d="M3 6h18"/><path d="M19 6l-1 14a2 2 0 0 1-2 2H8a2 2 0 0 1-2-2L5 6"/><path d="M8 6V4a2 2 0 0 1 2-2h4a2 2 0 0 1 2 2v2"/>',check:'<path d="M20 6 9 17l-5-5"/>',dl:'<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><path d="m7 10 5 5 5-5"/><path d="M12 15V3"/>',eye:'<path d="M2 12s4-8 10-8 10 8 10 8-4 8-10 8S2 12 2 12z"/><circle cx="12" cy="12" r="3"/>',layout:'<rect x="3" y="3" width="7" height="9"/><rect x="14" y="3" width="7" height="5"/><rect x="14" y="12" width="7" height="9"/><rect x="3" y="16" width="7" height="5"/>',search:'<circle cx="11" cy="11" r="7"/><path d="m21 21-4.3-4.3"/>',alert:'<path d="M12 9v4M12 17h.01"/><path d="M10.3 3.9 1.8 18a2 2 0 0 0 1.7 3h17a2 2 0 0 0 1.7-3L13.7 3.9a2 2 0 0 0-3.4 0z"/>',file:'<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><path d="M14 2v6h6"/>',send:'<path d="m22 2-7 20-4-9-9-4z"/><path d="M22 2 11 13"/>',msg:'<path d="M21 15a2 2 0 0 1-2 2H7l-4 4V5a2 2 0 0 1 2-2h14a2 2 0 0 1 2 2z"/>',save:'<path d="M19 21H5a2 2 0 0 1-2-2V5a2 2 0 0 1 2-2h11l5 5v11a2 2 0 0 1-2 2z"/><path d="M17 21v-8H7v8M7 3v5h8"/>',undo:'<path d="M9 14 4 9l5-5"/><path d="M4 9h10a6 6 0 0 1 0 12h-3"/>',clock:'<circle cx="12" cy="12" r="9"/><path d="M12 7v5l3 2"/>',coin:'<circle cx="12" cy="12" r="9"/><path d="M9 8h6M9 12h6M9 8c5 0 5 8 0 8h0"/>',hist:'<path d="M3 12a9 9 0 1 0 3-6.7L3 8"/><path d="M3 3v5h5M12 7v5l3 2"/>',tick:'<circle cx="12" cy="12" r="9"/><path d="m8 12 3 3 5-6"/>'};
const ic=n=>`<svg class="i" viewBox="0 0 24 24" aria-hidden="true">${IC[n]||""}</svg>`;
let askCb=null,onlyAv=false;
function ask(msg,cb,inp){askCb=cb;$("mod").innerHTML=`<div class="modal" onclick="if(event.target==this)closeM()"><div role="dialog"><p>${msg}</p>${inp?`<input id="askv" type="number" min="1" placeholder="Amount in Rs.">`:""}<p><button onclick="askOk()">${ic("check")}Confirm</button> <button class="ghost" onclick="closeM()">Cancel</button></p></div></div>`}
function askOk(){const x=$("askv"),val=x?x.value:null,cb=askCb;closeM();if(cb)cb(val)}
function borrow(id){const b=book(id);if(!me||me.role!="student")return toast("Log in as a student to borrow.");if(avail(b)<1)return toast("No copies are available right now.");const act=db.loans.filter(l=>l.userId==me.id&&!l.ret);if(act.length>=3)return toast("You can borrow up to 3 books at a time.");if(act.some(l=>l.bookId==id))return toast("You already have this book.");db.loans.push({id:db.nid++,bookId:id,userId:me.id,issue:d0(),due:addD(d0(),LOAN),ret:null,extra:0});save();closeM();toast("Borrowed. Due "+addD(d0(),LOAN));render()}
const toast=m=>{const t=$("toast");t.textContent=m;t.style.display="block";clearTimeout(toast.h);toast.h=setTimeout(()=>t.style.display="none",2600)};
function go(x,t){view=x;if(t!==undefined)tab=t;edit=null;render();scrollTo(0,0)}
function nav(){let h=`<div class="brand" onclick="go('home')">${ic("book")}Greenfield College Library</div><button class="ghost sm" onclick="go('home')">${ic("search")}Catalog</button><button class="ghost sm" onclick="go('contact')">${ic("mail")}Contact</button>`;
if(me)h+=`<button class="ghost sm" onclick="go('dash',me.role=='admin'?'books':'history')">${ic(me.role=="admin"?"layout":"user")}${me.role=="admin"?"Admin dashboard":"My account"}</button><button class="sm" onclick="logout()">${ic("logout")}Log out (${esc(me.name.split(" ")[0])})</button>`;
else h+=`<button class="ghost sm" onclick="go('login')">${ic("login")}Log in</button><button class="sm" onclick="go('register')">${ic("plus")}Register</button>`;
$("nav").innerHTML=h}
function render(){nav();$("app").innerHTML=({home,login:loginV,register:regV,contact,dash})[view]()}
function bookCards(){const s=q.toLowerCase(),L=db.books.filter(b=>(!cat||b.cat==cat)&&(!onlyAv||avail(b)>0)&&(!s||[b.title,b.author,b.isbn,b.cat].join(" ").toLowerCase().includes(s)));
return L.length?L.map(b=>{const a=avail(b);return`<div class="book" tabindex="0" onclick="openBook(${b.id})" onkeydown="if(event.key=='Enter')openBook(${b.id})"><h3>${esc(b.title)}</h3><small>${esc(b.author)}</small><span class="muted">${esc(b.cat)}</span><span class="tag ${a?"":"no"}">${a?a+" of "+b.total+" available":"All copies issued"}</span></div>`}).join(""):`<p class="muted">No books match ?${esc(q)}?. Try a title, author, ISBN or subject.</p>`}
function home(){const cats=[...new Set(db.books.map(b=>b.cat))];
return`<div class="hero"><h1>Find a book, check it out, keep track.</h1><p>Search the college collection by title, author, ISBN or subject. Log in to see your borrowed books, due dates and fines.</p></div>
<div class="row"><input id="q" placeholder="Search books" value="${esc(q)}" oninput="q=this.value;$('grid').innerHTML=bookCards()" aria-label="Search books"><select aria-label="Subject" onchange="cat=this.value;$('grid').innerHTML=bookCards()"><option value="">All subjects</option>${cats.map(c=>`<option ${c==cat?"selected":""}>${esc(c)}</option>`).join("")}</select></div>
```

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<p><label style="display:inline-flex;gap:8px;align-items:center;margin:0;color:var(--ink)"><input type="checkbox" style="width:auto" ${onlyAv?"checked":""} onchange="onlyAv=this.checked;$('grid').innerHTML=bookCards()">Show only available books</label></p><div class="grid" id="grid">${bookCards()}</div>`}
function openBook(id){const b=book(id),a=avail(b);const due=db.loans.filter(l=>l.bookId==id&&!l.ret).map(l=>l.due).sort()[0];
$("mod").innerHTML=`<div class="modal" onclick="if(event.target==this)closeM()"><div role="dialog" aria-label="Book details"><h2>${esc(b.title)}</h2><p>${esc(b.author)}</p><p>${esc(b.desc)}</p><table style="min-width:0"><tr><th>ISBN</th><td>${esc(b.isbn)}</td></tr><tr><th>Subject</th><td>${esc(b.cat)}</td></tr><tr><th>Copies</th><td>${a} of ${b.total} available</td></tr>${!a&&due?`<tr><th>Next return</th><td>${due}</td></tr>`:""}</table>${a?(me&&me.role=="student"?`<button onclick="borrow(${b.id})">${ic("book")}Borrow this book</button> `:me?`<p class="muted">Librarians issue books from Issue &amp;amp; return in the dashboard.</p>`:`<p class="muted">Log in as a student to borrow this book.</p>`):`<p class="muted">All copies are out. Check back after the next return date.</p>`}<button class="ghost" onclick="closeM()">Close</button></div></div>`}
const closeM=()=>$("mod").innerHTML="";
function loginV(){return`<div class="panel narrow"><h2>Log in</h2><label for="e">Email</label><input id="e" type="email" autocomplete="username"><label for="p">Password</label><input id="p" type="password" autocomplete="current-password" onkeydown="if(event.key=='Enter')login()"><p><button onclick="login()">${ic("login")}Log in</button></p><p class="muted">Demo accounts: admin@greenfield.edu / admin123 (librarian), priya@greenfield.edu / student123 (student).</p><p class="muted">New student? <a href="#" onclick="go('register');return false">Register here</a>.</p></div>`}
function login(){const u=db.users.find(u=>u.email==v("e").toLowerCase()&&u.pw==$("p").value);if(!u)return toast("Email or password is incorrect.");me=u;toast("Welcome, "+u.name.split(" ")[0]);go("dash",u.role=="admin"?"books":"history")}
function logout(){me=null;go("home");toast("Logged out")}
function regV(){return`<div class="panel narrow"><h2>Create student account</h2><label for="n">Full name</label><input id="n"><label for="e">Email</label><input id="e" type="email"><label for="r">Roll number</label><input id="r"><div class="row"><div><label for="y">Year</label><select id="y">${[1,2,3,4].map(i=>`<option>${i}</option>`).join("")}</select></div><div><label for="s">Semester</label><select id="s">${[1,2,3,4,5,6,7,8].map(i=>`<option>${i}</option>`).join("")}</select></div></div><label for="dp">Department</label><input id="dp"><label for="p">Password (6+ characters)</label><input id="p" type="password"><p><button onclick="register()">${ic("plus")}Create account</button></p></div>`}
function register(){const e=v("e").toLowerCase();if(!v("n")||!e||!v("r"))return toast("Enter name, email and roll number.");if($("p").value.length<6)return toast("Password needs at least 6 characters.");if(db.users.some(u=>u.email==e))return toast("That email is already registered.");
const u={id:db.nid++,name:v("n"),email:e,pw:$("p").value,role:"student",roll:v("r"),dept:v("dp"),year:+$("y").value,sem:+$("s").value};db.users.push(u);save();me=u;toast("Account created");go("dash","history")}
function contact(){return`<div class="panel narrow"><h2>Contact the library</h2><p class="muted">Open Mon?Sat, 9 am to 6 pm. Ask about a book, a fine or study material.</p><label for="cn">Name</label><input id="cn" value="${esc(me?me.name:"")}"><label for="ce">Email</label><input id="ce" type="email" value="${esc(me?me.email:"")}"><label for="cm">Message</label><textarea id="cm" rows="4"></textarea><p><button onclick="sendMsg()">${ic("send")}Send message</button></p></div>`}
function sendMsg(){if(!v("cn")||!v("ce")||!v("cm"))return toast("Fill in all three fields.");db.msgs.push({id:db.nid++,name:v("cn"),email:v("ce"),text:v("cm"),date:d0()});save();toast("Message sent. We?ll reply by email.");go("home")}
function dash(){if(!me)return loginV();return me.role=="admin"?adminV():studentV()}
const tabs=(L)=>`<div class="tabs">${L.map(([k,n,i])=>`<button class="${tab==k?"on":""}" onclick="go('dash','${k}')">${ic(i)}${n}</button>`).join("")}</div>`;
/* ---------- student ---------- */
function studentV(){const mine=db.loans.filter(l=>l.userId==me.id),tot=mine.reduce((s,l)=>s+fine(l),0),act=mine.filter(l=>!l.ret).length;
let h=`<h2>Hello, ${esc(me.name)}</h2><div class="stats"><div class="stat">${ic("book")}<b>${act}</b>Books borrowed now</div><div class="stat">${ic("coin")}<b>Rs.${tot}</b>Fine due</div><div class="stat">${ic("hist")}<b>${mine.length}</b>Books borrowed in total</div></div>`+tabs([["history","Borrow history","hist"],["mat","Study material","file"],["prof","My profile","user"]]);
if(tab=="history")h+=`<div class="panel tw"><table><tr><th>Book</th><th>Issue date</th><th>Due date</th><th>Status</th><th>Fine</th></tr>${mine.length?mine.slice().reverse().map(l=>{const f=fine(l),late=!l.ret&&l.due<d0();return`<tr><td>${esc(book(l.bookId).title)}</td><td>${l.issue}</td><td>${l.due}</td><td>${l.ret?"Returned "+l.ret:late?`<span class="late">Overdue by ${diff(l.due,d0())} days</span>`:"Borrowed"}</td><td>${f?`<span class="late">Rs.${f}</span>`:"?"}</td></tr>`}).join(""):`<tr><td colspan="5" class="muted">You haven?t borrowed anything yet. Browse the catalog and ask the librarian to issue a book.</td></tr>`}</table><p class="muted">Loans last ${LOAN} days. Late returns cost Rs.${RATE} per day.</p></div>`;
if(tab=="mat")h+=matFilter()+`<div class="panel tw" id="mats">${matTable(false)}</div>`;
if(tab=="prof")h+=`<div class="panel"><h3>My profile</h3><div class="row"><div><label>Name</label><input id="pn" value="${esc(me.name)}"></div><div><label>Email</label><input id="pe" value="${esc(me.email)}"></div></div><div class="row"><div><label>Roll number</label><input id="pr" value="${esc(me.roll)}"></div><div><label>Department</label><input id="pd" value="${esc(me.dept)}"></div></div><div class="row"><div><label>Year</label><input id="py" type="number" min="1" max="4"
```

```
value="${me.year}"></div><div><label>Semester</label><input id="ps" type="number" min="1" max="8"
value="${me.sem}"></div></div><p><button onclick="saveProf()">${ic("save")}Save profile</button></p></div>`;
return h}
function saveProf(){if(!v("pn")||!v("pe"))return toast("Name and email are required.");Object.assign(me,{name:v("pn"),email:v("pe")
.toLowerCase(),roll:v("pr"),dept:v("pd"),year:+v("py"),sem:+v("ps")});save();nav();toast("Profile saved")}
/* ---------- materials ---------- */
function matFilter(){return`<div class="panel"><div class="row"><div><label>Search</label><input id="mt" value="${esc(mq.t)}"
oninput="mq.t=this.value;$('mats').innerHTML=matTable(${me.role=="admin"})"></div><div><label>Year</label><select
id="my" onchange="mq.y=this.value;$('mats').innerHTML=matTable(${me.role=="admin"})"><option
value="">All</option>${[1,2,3,4].map(i=>`<option
${mq.y==i?"selected":""}>${i}</option>`).join("")}</select></div><div><label>Semester</label><select id="ms"
onchange="mq.s=this.value;$('mats').innerHTML=matTable(${me.role=="admin"})"><option
value="">All</option>${[1,2,3,4,5,6,7,8].map(i=>`<option
${mq.s==i?"selected":""}>${i}</option>`).join("")}</select></div></div></div>`}
function matTable(adm){const L=db.mats.filter(m=>(!mq.y||m.year==mq.y)&&(!mq.s||m.sem==mq.s)&&(!mq.t||m.title.toLowerCas
e().includes(mq.t.toLowerCase())));
return`<table><tr><th>Title</th><th>Year</th><th>Sem</th><th></th></tr>${L.length?L.map(m=>`<tr><td>${esc(m.title)}</td>
<td>${m.year}</td><td>${m.sem}</td><td><button class="sm ghost"
onclick="viewMat(${m.id})">${ic("eye")}View</button> <button class="sm"
onclick="dlMat(${m.id})">${ic("dl")}Download</button>${adm?` <button class="sm ghost"
onclick="editMat(${m.id})">${ic("edit")}Edit</button> <button class="sm del"
onclick="delMat(${m.id})">${ic("trash")}Delete</button>`:""}</td></tr>`).join(""):`<tr><td colspan="4" class="muted">No
material for this year and semester yet.</td></tr>`}</table>`}
function viewMat(id){const m=db.mats.find(x=>x.id==id);$("mod").innerHTML=`<div class="modal"
onclick="if(event.target==this)closeM()"><div><h2>${esc(m.title)}</h2><p class="muted">Year ${m.year}, semester
${m.sem}</p><p>${esc(m.body)}</p><button onclick="closeM()">Close</button></div></div>`}
async function dlMat(id){const m=db.mats.find(x=>x.id==id);try{const dl=await claude.use("downloads");if(dl){await
dl.save({filename:m.title.replace(/\W+/g,"_")+".txt",data:m.title+"\nYear "+m.year+", Semester
"+m.sem+"\n\n"+m.body});return toast("Download ready")}}catch(e){if(e&&e.code=="declined")return}try{const
a=document.createElement("a");a.href=URL.createObjectURL(new Blob([m.title+"\nYear "+m.year+", Semester
"+m.sem+"\n\n"+m.body],{type:"text/plain"}));a.download=m.title.replace(/\W+/g,"_")+".txt";a.click();toast("Download
started")}catch(e){toast("Download isn?t available here. Use View instead.")}}
/* ---------- admin ---------- */
function adminV(){const act=db.loans.filter(l=>!l.ret),od=act.filter(l=>l.due<d0()).length;
let h=`<h2>Admin dashboard</h2><div class="stats"><div
class="stat">${ic("book")}<b>${db.books.reduce((s,b)=>s+b.total,0)}</b>Total copies</div><div
class="stat">${ic("tick")}<b>${db.books.reduce((s,b)=>s+avail(b),0)}</b>Available now</div><div
class="stat">${ic("clock")}<b>${act.length}</b>Issued now</div><div
class="stat">${ic("alert")}<b>${od}</b>Overdue</div><div class="stat">${ic("users")}<b>${db.users.filter(u=>u.role=="stud
ent").length}</b>Students</div></div>`+tabs([["books","Books","book"],["issue","Issue & return","undo"],["mat","Study
material","file"],["users","Students","users"],["msgs","Messages","msg"]]);
if(tab=="books"){const b=edit&&edit.t=="b"?book(edit.id):{};
h+=`<div class="panel"><h3>${b.id?"Update book":"Add a book"}</h3><div class="row"><div><label>Title</label><input
id="bt" value="${esc(b.title)}"></div><div><label>Author</label><input id="ba" value="${esc(b.author)}"></div></div><div
class="row"><div><label>ISBN</label><input id="bi" value="${esc(b.isbn)}"></div><div><label>Subject</label><input
id="bc" value="${esc(b.cat)}"></div><div><label>Copies</label><input id="bn" type="number" min="1"
value="${b.total||1}"></div></div><label>Description</label><textarea id="bd"
rows="2">${esc(b.desc)}</textarea><p><button onclick="saveBook(${b.id||0})">${ic(b.id?"save":"plus")}${b.id?"Save
changes":"Add book"}</button> ${b.id?`<button class="ghost" onclick="go('dash','books')">Cancel</button>`:""}</p></div>
<div class="panel tw"><table><tr><th>Title</th><th>Author</th><th>Subject</th><th>Available</th><th></th></tr>${db.books.
map(x=>`<tr><td>${esc(x.title)}</td><td>${esc(x.author)}</td><td>${esc(x.cat)}</td><td><span class="tag
${avail(x)?"":"no"}">${avail(x)?"Available "+avail(x)+" / "+x.total:"All issued"}</span></td><td><button class="sm ghost"
onclick="edit={t:'b',id:${x.id}};render();scrollTo(0,0)">${ic("edit")}Edit</button> <button class="sm del"
onclick="delBook(${x.id})">${ic("trash")}Delete</button></td></tr>`).join("")}</table></div>`}
if(tab=="issue"){const st=db.users.filter(u=>u.role=="student");
h+=`<div class="panel"><h3>Issue a book</h3><div class="row"><div><label>Student</label><select
id="is">${st.map(u=>`<option value="${u.id}">${esc(u.name)}
(${esc(u.roll)})</option>`).join("")}</select></div><div><label>Book</label><select
id="ib">${db.books.filter(b=>avail(b)>0).map(b=>`<option
value="${b.id}">${esc(b.title)}</option>`).join("")}</select></div><div><label>Days</label><input id="id" type="number"
min="1" value="${LOAN}"></div></div><p><button onclick="issue()">${ic("clock")}Issue book</button></p></div>
<div class="panel tw"><h3>Loans</h3><table><tr><th>Student</th><th>Book</th><th>Issued</th><th>Due</th><th>Status</th
><th>Fine</th><th></th></tr>${db.loans.slice().reverse().map(l=>{const f=fine(l);return`<tr><td>${esc(user(l.userId).name)}<
/td><td>${esc(book(l.bookId).title)}</td><td>${l.issue}</td><td>${l.due}</td><td>${l.ret?"Returned
"+l.ret:l.due<d0()?`<span class="late">Overdue</span>`:"Issued"}</td><td>${f?`<span
```

```
class="late">Rs.${f}</span>`:"?"}</td><td>${l.ret?"":`<button class="sm"
onclick="ret(${l.id})">${ic("undo")}Return</button> `}<button class="sm ghost" onclick="addFine(${l.id})">${ic("coin")}Add
fine</button></td></tr>`}).join("")}</table></div>`}
if(tab=="mat"){const m=edit&&edit.t=="m"?db.mats.find(x=>x.id==edit.id):{};
h+=`<div class="panel"><h3>${m.id?"Update material":"Add material"}</h3><div class="row"><div
style="flex:2"><label>Title</label><input id="xt" value="${esc(m.title)}"></div><div><label>Year</label><select
id="xy">${[1,2,3,4].map(i=>`<option
${m.year==i?"selected":""}>${i}</option>`).join("")}</select></div><div><label>Semester</label><select
id="xs">${[1,2,3,4,5,6,7,8].map(i=>`<option
${m.sem==i?"selected":""}>${i}</option>`).join("")}</select></div></div><label>Content</label><textarea id="xb"
rows="3">${esc(m.body)}</textarea><p><button onclick="saveMat(${m.id||0})">${ic(m.id?"save":"plus")}${m.id?"Save
changes":"Add material"}</button></p></div>`+matFilter()+`<div class="panel tw" id="mats">${matTable(true)}</div>`}
if(tab=="users")h+=`<div class="panel tw"><table><tr><th>Name</th><th>Roll</th><th>Department</th><th>Year /
Sem</th><th>Borrowed</th><th>Fine due</th></tr>${db.users.filter(u=>u.role=="student").map(u=>{const
L=db.loans.filter(l=>l.userId==u.id);const f=L.reduce((s,l)=>s+fine(l),0);return`<tr><td>${esc(u.name)}<br><small
class="muted">${esc(u.email)}</small></td><td>${esc(u.roll)}</td><td>${esc(u.dept)}</td><td>${u.year} /
${u.sem}</td><td>${L.filter(l=>!l.ret).length}</td><td>${f?`<span
class="late">Rs.${f}</span>`:"?"}</td></tr>`}).join("")}</table></div>`;
if(tab=="msgs")h+=`<div class="panel">${db.msgs.length?db.msgs.slice().reverse().map(m=>`<p><b>${esc(m.name)}</b> <span
class="muted">${esc(m.email)}, ${m.date}</span><br>${esc(m.text)}</p>`).join("<hr>"):`<p class="muted">No messages yet.
Contact form submissions show up here.</p>`}</div>`;
return h}
function saveBook(id){if(!v("bt")||!v("ba"))return toast("Title and author are required.");const
o={title:v("bt"),author:v("ba"),isbn:v("bi"),cat:v("bc")||"General",total:Math.max(1,+v("bn")||1),desc:v("bd")};
if(id)Object.assign(book(id),o);else db.books.push({id:db.nid++,...o});save();toast(id?"Book updated":"Book
added");go("dash","books")}
function delBook(id){if(db.loans.some(l=>l.bookId==id&&!l.ret))return toast("This book is issued. Return it before
deleting.");ask("Delete this book from the catalog?",()=>{db.books=db.books.filter(b=>b.id!=id);save();toast("Book
deleted");render()})}
function issue(){const b=+v("ib"),u=+v("is");if(!b||!u)return toast("Choose a student and a book.");const
n=+v("id")||LOAN;db.loans.push({id:db.nid++,bookId:b,userId:u,issue:d0(),due:addD(d0(),n),ret:null,extra:0});save();toast("Book
issued. Due "+addD(d0(),n));render()}
function ret(id){const l=db.loans.find(x=>x.id==id);l.ret=d0();save();const f=fine(l);toast(f?"Returned. Fine due: Rs."+f:"Returned
on time");render()}
function addFine(id){ask("Fine amount in Rs. (for damage or loss):",a=>{const n=Math.round(+a);if(!(n>0))return toast("Enter an
amount above 0.");db.loans.find(x=>x.id==id).extra+=n;save();toast("Fine of Rs."+n+" added");render()},true)}
function saveMat(id){if(!v("xt"))return toast("Enter a title.");const
o={title:v("xt"),year:+v("xy"),sem:+v("xs"),body:v("xb")};if(id)Object.assign(db.mats.find(m=>m.id==id),o);else
db.mats.push({id:db.nid++,...o});save();toast(id?"Material updated":"Material added");go("dash","mat")}
function editMat(id){edit={t:"m",id};render();scrollTo(0,0)}
function delMat(id){ask("Delete this material?",()=>{db.mats=db.mats.filter(m=>m.id!=id);save();toast("Material
deleted");render()})}
render();
</script></body></html>
```

Output

The library website opens with a searchable catalog showing availability. The librarian logs in to add, update or delete books, issue and return them, add fines and manage study material. The student logs in to see borrow history, due dates, fines and study material.

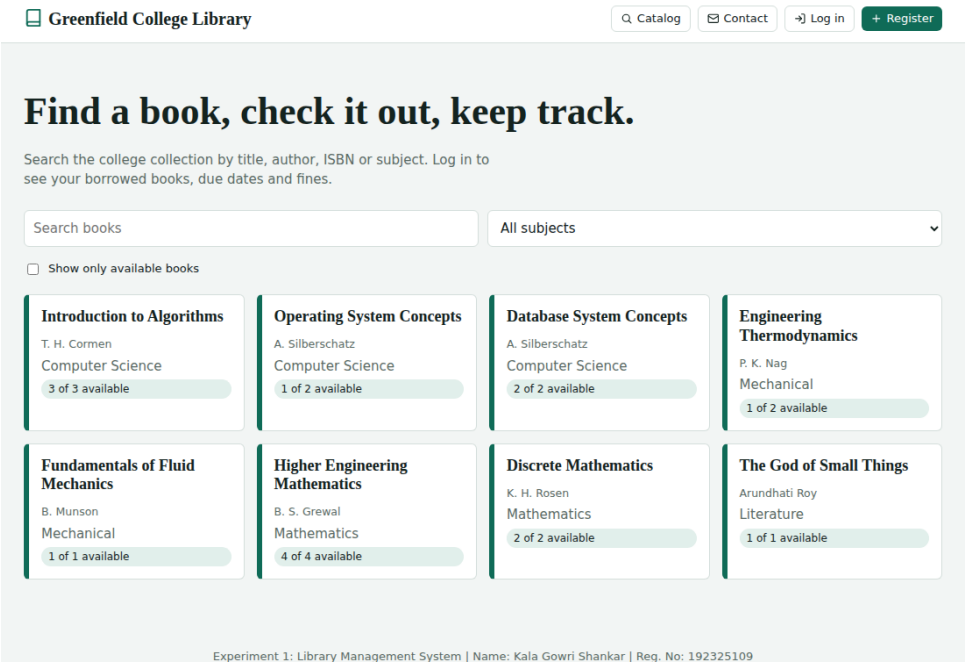


Figure 1: Catalog with search and availability

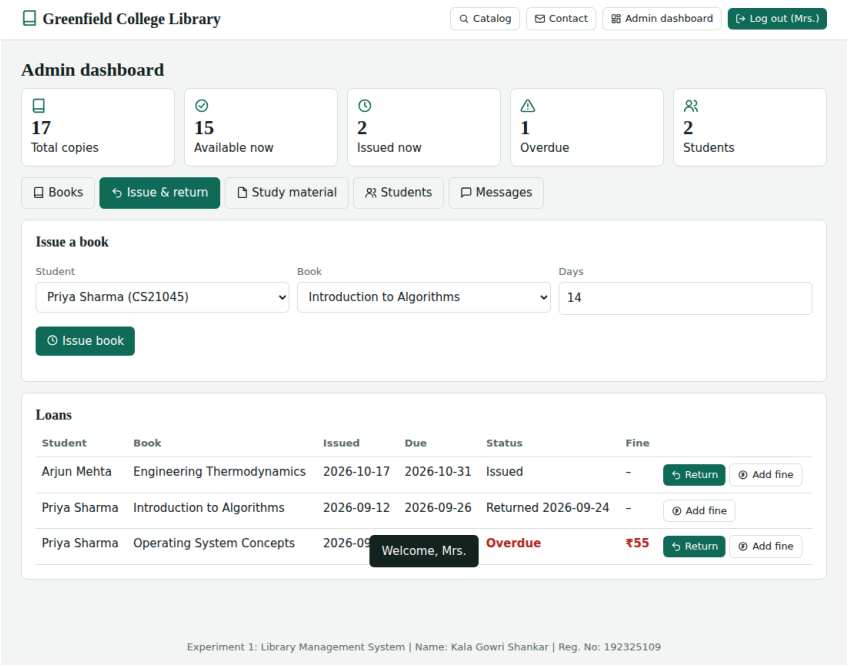


Figure 2: Librarian dashboard: issue and return books

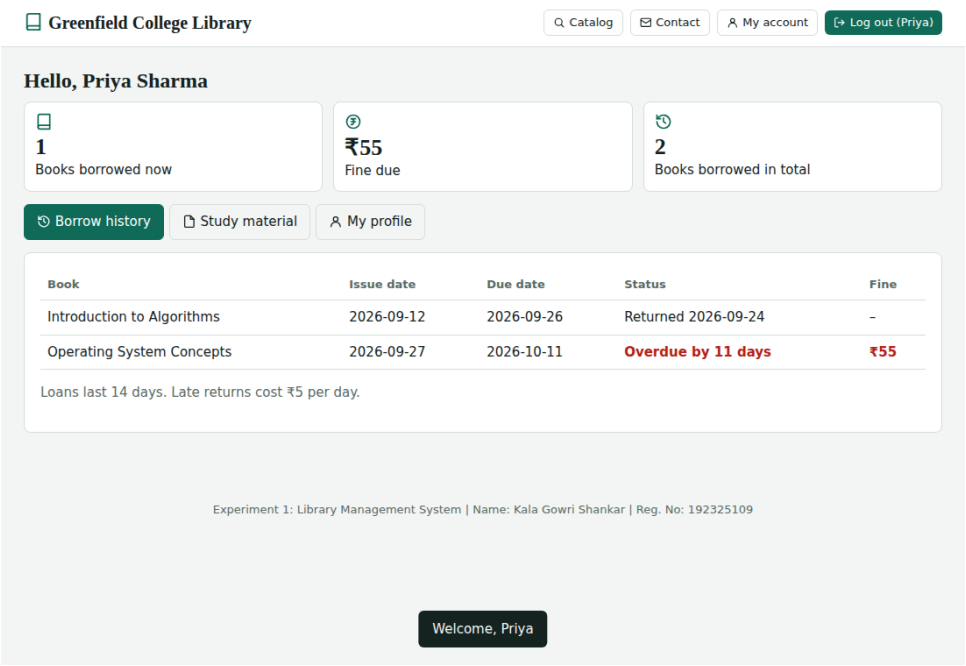


Figure 3: Student account: borrow history and fine

Result

Thus, a library management website with catalog search, user login, borrowing and return tracking, fine calculation and study material was created successfully.

Experiment 2: Online Hotel Room Booking System

Aim

To create a hotel room booking website where users register, check room availability for chosen dates and guests, book and pay, view, change or cancel bookings up to 3 days before arrival, and where the administrator manages room quantity, prices, bookings and static pages.

HTML Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Grand Stay Hotel Booking</title>
<style>
body{font-family:Arial;background:#fff7ee;margin:0;color:#49352a}
header{background:#68452f;color:#fff;padding:16px;display:flex;flex-wrap:wrap;gap:8px;align-items:center}
header h1{font-size:22px;margin:0 auto 0 0}
button{background:#95613e;color:#fff;border:0;border-radius:6px;padding:8px 12px;cursor:pointer;margin:2px}
nav button.o{background:transparent;border:1px solid #fff}
button.x{background:#b3261e}
main{max-width:920px;margin:20px auto;background:#fff;padding:20px;border-radius:12px}
input,select,textarea{padding:9px;border:1px solid #d9c2ae;border-radius:6px;box-sizing:border-box;margin:4px 0;width:100%}
.room{border:1px solid #e1cbb8;border-radius:8px;padding:12px;margin:10px
    0;display:flex;gap:10px;flex-wrap:wrap;align-items:center;justify-content:space-between}
.tw{overflow-x:auto}table{width:100%;border-collapse:collapse}th,td{border:1px solid
    #e1cbb8;padding:8px;text-align:left}th{background:#f3e5d8}
.ok{color:#176b4d;font-weight:bold}.err{color:#b3261e;font-weight:bold}
.row{display:flex;gap:8px;flex-wrap:wrap}.row>*{flex:1;min-width:130px}
td input{min-width:110px}
</style></head><body>
<header><h1>Grand Stay Hotel</h1><nav id="nav"></nav></header>
<main id="app"></main><p id="msg" style="text-align:center"></p>
<script>
const $=i=>document.getElementById(i),V=i=>$(i).value.trim(),K="hotel_lab";
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
const fmt=d=>d.getFullYear()+"-"+String(d.getMonth()+1).padStart(2,"0")+"-"+String(d.getDate()).padStart(2,"0");
const today=()=>fmt(new Date());
const add=n=>{const d=new Date();d.setDate(d.getDate()+n);return fmt(d)};
const days=(a,b)=>Math.round((new Date(b)-new Date(a))/864e5);
function seed(){return{id:10,
users:[{id:1,name:"Admin",email:"admin@hotel.com",pw:"admin123",admin:1,phone:""},{id:2,name:"Kala",email:"kala@gmail.co
    m",pw:"1234",phone:"9876543210"}],
rooms:[{t:"Standard",qty:5,price:1800,cap:2},{t:"Deluxe",qty:4,price:2800,cap:3},{t:"Family
    Suite",qty:3,price:4200,cap:5},{t:"Presidential",qty:1,price:7500,cap:6}],
bookings:[{id:1,uid:2,room:"Deluxe",in:add(6),out:add(8),g:2,total:5600,st:"Booked"}],
pages:{about:"Grand Stay Hotel offers comfortable rooms in the heart of the city with free Wi-Fi and breakfast.",
rooms:"Standard: one double bed. Deluxe: city view. Family Suite: two beds and living area. Presidential: premium suite.",
contact:"Phone: +91 98765 43210 | Email: help@grandstay.com | Address: 12 Beach Road, Chennai.",
faq:"Q: Can I cancel? A: Yes, up to 3 days before arrival. Q: Check-in time? A: 12 noon.",
travel:"Airport is 12 km away. Local taxis and metro are available near the hotel.",
privacy:"We keep your details private and use them only for bookings."}}}
let S;try{S=JSON.parse(localStorage.getItem(K))}catch(e){}S=S||seed();
const save=()=>{try{localStorage.setItem(K,JSON.stringify(S))}catch(e){}};
let me=null,view="home",arg=null,q={in:"",out:"",g:1};
const say=(m,bad)=>{$("msg").className=bad?"err":"ok";$("msg").textContent=m};
const avail=(t,a,b,skip)=>S.rooms.find(x=>x.t==t).qty-S.bookings.filter(x=>x.st=="Booked"&&x.room==t&&x.id!=skip&&x.in<b
    &&a<x.out).length;
const go=(v,a)=>{view=v;arg=a;say("");render()};
function render(){
```

```
$("nav").innerHTML=`<button class="o" onclick="go('home')">Home</button><button class="o" onclick="go('info','about')">Hotel Info</button>`+
(me?(me.admin?`<button class="o" onclick="go('arooms')">Rooms</button><button class="o" onclick="go('abook')">Bookings</button><button class="o" onclick="go('apages')">Pages</button>`:`<button class="o" onclick="go('my')">My Bookings</button><button class="o" onclick="go('prof')">Profile</button>`)+`<button onclick="me=null;go('home')">Logout (${esc(me.name)})</button>`:`<button onclick="go('login')">Login</button><button onclick="go('reg')">Register</button>`);
$("app").innerHTML=VIEW[view]()}
function list(){return S.rooms.map(r=>{const a=q.in?avail(r.t,q.in,q.out):null,ok=a===null||(a>0&&q.g<=r.cap);
return`<div class="room"><div><b>${r.t}</b> - Rs.${r.price}/night<br>Sleeps ${r.cap} | Wi-Fi | Breakfast${a!==null?`<br><span class="${ok?"ok":"err"}">${a>0?a+" room(s) available":"Sold out"}${q.g>r.cap?" (too many guests)":""}</span>`:""}</div>
<button ${ok&&q.in?"":"disabled"} onclick="pick('${r.t}')">Book Now</button></div>`}).join("")}
function search(){const a=V("a"),b=V("b"),g=+V("g");if(!a||!b||a<today()||b<=a)return say("Choose valid check-in and check-out dates.",1);q={in:a,out:b,g};render()}
function pick(t){if(!me){go("login");return say("Please login to book a room.",1)}go("book",t)}
const VIEW={
home:()=>`<h2>Search Rooms</h2><div class="row"><div>Check-in<input id="a" type="date" min="${today()}" value="${q.in}"></div><div>Check-out<input id="b" type="date" value="${q.out}"></div><div>Guests<input id="g" type="number" min="1" max="8" value="${q.g}"></div></div><button onclick="search()">Check Availability</button>${list()}`,
info:()=>`<h2>Hotel Information</h2>${Object.keys(S.pages).map(k=>`<button onclick="go('info','${k}')">${k=="faq"?"Customer Q&A":k=="travel"?"Travel Guide":k=="rooms"?"Room Types":k[0].toUpperCase()+k.slice(1)}</button>`).join("")}<p>${esc(S.pages[arg])}</p>`,
login:()=>`<h2>Login</h2><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="login()">Login</button><p>Demo: kala@gmail.com / 1234, admin@hotel.com / admin123</p>`,
reg:()=>`<h2>Register</h2><input id="n" placeholder="Full name"><input id="e" type="email" placeholder="Email"><input id="ph" placeholder="Phone"><input id="p" type="password" placeholder="Password"><button onclick="reg()">Create Account</button>`,
book:()=>{const r=S.rooms.find(x=>x.t==arg),n=days(q.in,q.out);return`<h2>Book ${r.t}</h2><p>${q.in} to ${q.out} | ${n} night(s) | ${q.g} guest(s)<br><b>Total: Rs.${n*r.price}</b></p><input id="n" value="${esc(me.name)}"><input id="card" placeholder="Card number (demo payment gateway)" maxlength="16"><button onclick="pay('${r.t}',${n*r.price})">Pay and Confirm Booking</button>`},
my:()=>{const L=S.bookings.filter(b=>b.uid==me.id);return`<h2>My Booking History</h2><div class="tw"><table><tr><th>ID</th><th>Room</th><th>Check-in</th><th>Check-out</th><th>Total</th><th>Status</th><th></th></tr>${L.map(b=>{const can=b.st=="Booked"&&days(today(),b.in)>=3;
return arg==b.id?`<tr><td>${b.id}</td><td>${b.room}</td><td><input id="ei" type="date" value="${b.in}"></td><td><input id="eo" type="date" value="${b.out}"></td><td></td><td>${b.st}</td><td><button onclick="chg(${b.id})">Save</button></td></tr>`:
`<tr><td>${b.id}</td><td>${b.room}</td><td>${b.in}</td><td>${b.out}</td><td>Rs.${b.total}</td><td>${b.st}</td><td>${can?`<button onclick="go('my',${b.id})">Change</button><button class="x" onclick="cancel(${b.id})">Cancel</button>`:"-"}</td></tr>`}).join("")||`<tr><td colspan="7">No bookings yet.</td></tr>`}</table></div><p>Changes and cancellations are allowed up to 3 days before arrival.</p>`},
prof:()=>`<h2>My Profile</h2>Name<input id="n" value="${esc(me.name)}">Email<input value="${esc(me.email)}" disabled>Phone<input id="ph" value="${esc(me.phone)}">Password<input id="p" type="password" value="${esc(me.pw)}"><button onclick="saveProf()">Save Profile</button>`,
arooms:()=>`<h2>Manage Rooms</h2><div class="tw"><table><tr><th>Type</th><th>Quantity</th><th>Price (Rs.)</th><th></th></tr>${S.rooms.map((r,i)=>`<tr><td>${r.t}</td><td><input id="q${i}" type="number" min="0" value="${r.qty}"></td><td><input id="pr${i}" type="number" min="1" value="${r.price}"></td><td><button onclick="saveRoom(${i})">Update</button></td></tr>`).join("")}</table></div>`,
abook:()=>`<h2>All Bookings</h2><div class="tw"><table><tr><th>ID</th><th>Guest</th><th>Room</th><th>Check-in</th><th>Check-out</th><th>Status</th><th></th></tr>${S.bookings.map(b=>`<tr><td>${b.id}</td><td>${esc(S.users.find(u=>u.id==b.uid).name)}</td><td><select id="r${b.id}">${S.rooms.map(r=>`<option ${r.t==b.room?"selected":""}>${r.t}</option>`).join("")}</select></td><td><input id="i${b.id}" type="date" value="${b.in}"></td><td><input id="o${b.id}" type="date" value="${b.out}"></td><td>${b.st}</td><td><button onclick="aEdit(${b.id})">Save</button>${b.st=="Booked"?`<button class="x" onclick="cancel(${b.id})">Cancel</button>`:""}</td></tr>`).join("")}</table></div>`,
apages:()=>`<h2>Edit Static Pages</h2>${Object.keys(S.pages).map(k=>`<b>${k}</b><textarea id="pg_${k}" rows="2">${esc(S.pages[k])}</textarea>`).join("")}<button onclick="savePages()">Save All Pages</button>`};
function login(){const u=S.users.find(u=>u.email==V("e").toLowerCase()&&u.pw==$("p").value);if(!u)return say("Wrong email or password.",1);me=u;go(u.admin?"arooms":"home");say("Welcome, "+u.name)}
function reg(){const e=V("e").toLowerCase();if(!V("n")||!e||!$("p").value)return say("Fill name, email and password.",1);if(S.users.some(u=>u.email==e))return say("Email already registered.",1);
const u={id:S.id++,name:V("n"),email:e,phone:V("ph"),pw:$("p").value};S.users.push(u);save();me=u;go("home");say("Registered successfully.")}
```

```
function pay(t,total){if(V("card").length<12)return say("Enter a valid 12-16 digit card number.",1);
if(avail(t,q.in,q.out)<1)return say("Room no longer available.",1);
S.bookings.push({id:S.id++,uid:me.id,room:t,in:q.in,out:q.out,g:q.g,total,st:"Booked"});save();go("my");say("Payment successful. Confirmation email sent to "+me.email)}
function chg(id){const b=S.bookings.find(x=>x.id==id),a=V("ei"),o=V("eo");if(!a||o<=a||days(today(),a)<3)return say("Choose valid dates at least 3 days from today.",1);
if(avail(b.room,a,o,id)<1)return say("Room not available on those dates.",1);b.in=a;b.out=o;b.total=days(a,o)*S.rooms.find(r=>r.t==b.room).price;save();go("my");say("Booking changed.")}
function cancel(id){const b=S.bookings.find(x=>x.id==id);if(!me.admin&&days(today(),b.in)<3)return say("Cannot cancel within 3 days of arrival.",1);b.st="Cancelled";save();render();say("Booking cancelled.")}
function saveProf(){me.name=V("n");me.phone=V("ph");me.pw=$("p").value;save();render();say("Profile updated.")}
function saveRoom(i){S.rooms[i].qty=+V("q"+i);S.rooms[i].price=+V("pr"+i);save();say(S.rooms[i].t+" updated.")}
function aEdit(id){const b=S.bookings.find(x=>x.id==id),a=V("i"+id),o=V("o"+id);if(!a||o<=a)return say("Invalid dates.",1);b.room=V("r"+id);b.in=a;b.out=o;b.total=days(a,o)*S.rooms.find(r=>r.t==b.room).price;save();say("Booking "+id+" updated.")}
function savePages(){Object.keys(S.pages).forEach(k=>S.pages[k]=V("pg_"+k));save();say("Pages updated.")}
render();
</script></body></html>
```

Output

The hotel website shows four room types with live availability. A logged-in user books a room, pays through a demo gateway and sees the booking history with Change and Cancel buttons. The administrator edits or cancels any booking, changes room quantity and price, and edits the static pages.

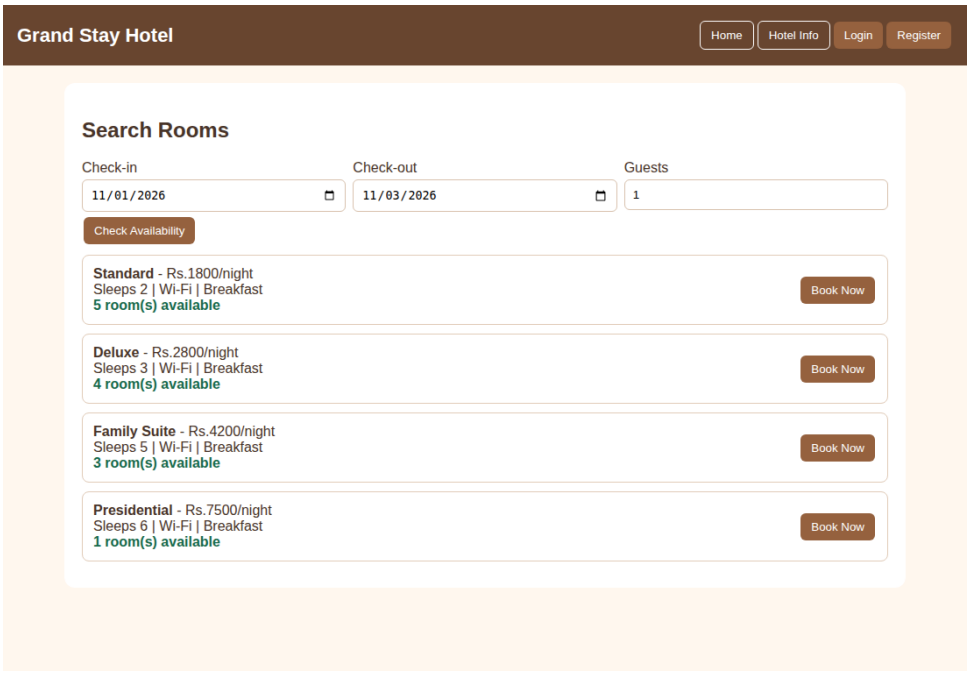


Figure 1: Room search with availability

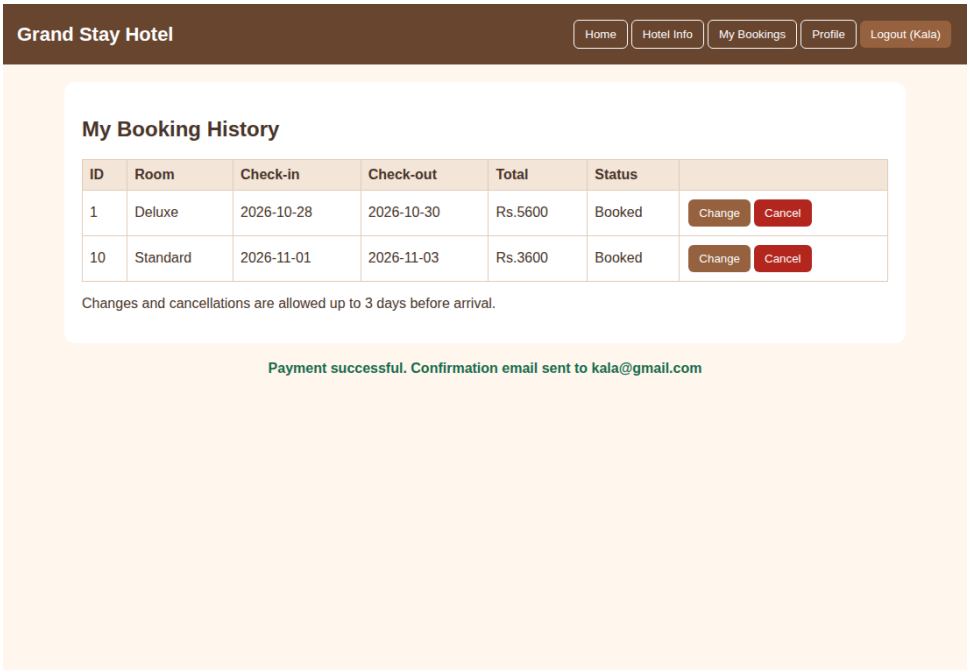


Figure 2: User booking history

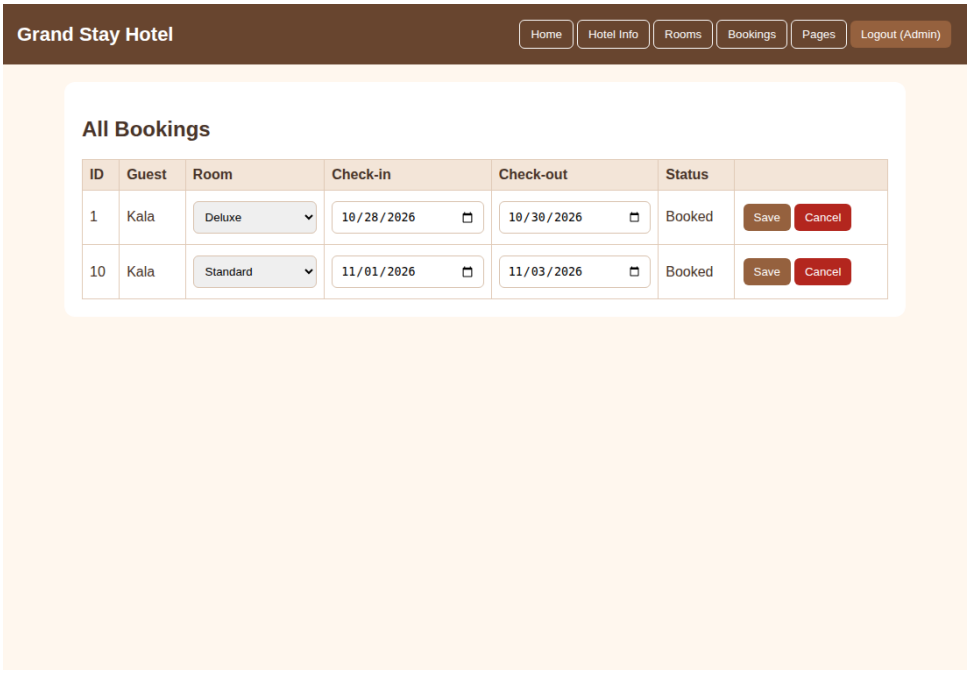


Figure 3: Administrator: all bookings

Result

Thus, an online hotel room booking website with availability check, booking management and an admin panel was created successfully.

Experiment 3: Online Bus Ticket Booking System

Aim

To develop a bus ticket booking website with login, registration, change and forgot password, bus search by route and date, seat selection, passenger details, ticket cancellation, comments, and an admin panel for bus categories, bus information, users and reports.

HTML Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>GoBus Ticket Booking</title>
<style>
body{font-family:Arial;background:#eef6f6;margin:0;color:#203b43}
header{background:#176b70;color:#fff;padding:16px;display:flex;flex-wrap:wrap;gap:8px;align-items:center}
header h1{font-size:22px;margin:0 auto 0 0}
button{background:#176b70;color:#fff;border:0;border-radius:6px;padding:8px 12px;cursor:pointer;margin:2px}
nav button{background:transparent;border:1px solid #fff}nav button.s{background:#f5b942;color:#203b43;border-color:#f5b942}
button.x{background:#b3261e}
main{max-width:920px;margin:20px auto;background:#fff;padding:20px;border-radius:12px}
input,select,textarea{padding:9px;border:1px solid #b9d3d5;border-radius:6px;box-sizing:border-box;margin:4px 0;width:100%}
.bus{border:1px solid #cfe2e3;border-radius:8px;padding:12px;margin:10px
    0;display:flex;gap:10px;flex-wrap:wrap;align-items:center;justify-content:space-between}
.seats{display:grid;grid-template-columns:repeat(5,52px);gap:6px;margin:10px 0}
.seats button{margin:0;padding:10px 0;background:#e3f7ee;color:#203b43;border:1px solid #176b70}
.seats button.sel{background:#176b70;color:#fff}.seats button:disabled{background:#ccc;border-color:#aaa;color:#777}
.tw{overflow-x:auto}table{width:100%;border-collapse:collapse}th,td{border:1px solid
    #cfe2e3;padding:8px;text-align:left}th{background:#dff0f1}
.ok{color:#176b4d;font-weight:bold}.err{color:#b3261e;font-weight:bold}
.row{display:flex;gap:8px;flex-wrap:wrap}.row>*{flex:1;min-width:130px}
</style></head><body>
<header><h1>GoBus Ticket Booking</h1><nav id="nav"></nav></header>
<main id="app"></main><p id="msg" style="text-align:center"></p>
<script>
const $=i=>document.getElementById(i),V=i=>$(i).value.trim(),K="bus_lab";
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
const fmt=d=>d.getFullYear()+"-"+String(d.getMonth()+1).padStart(2,"0")+"-"+String(d.getDate()).padStart(2,"0");
const today=()=>fmt(new Date());
function seed(){return{id:10,cats:["Volvo","Sleeper Coach","Coach"],
users:[{id:1,name:"Admin",email:"admin@gobus.com",pw:"admin123",admin:1},{id:2,name:"Kala",email:"kala@gmail.com",pw:"
    1234"}],
buses:[{id:1,name:"GreenLine Express",cat:"Volvo",from:"Chennai",to:"Bengaluru",time:"21:30",price:850,seats:20,desc:"AC
    Volvo with charging points and pushback seats."},
{id:2,name:"City Rider",cat:"Coach",from:"Chennai",to:"Bengaluru",time:"08:00",price:450,seats:20,desc:"Non-AC coach, daily
    morning service."},
{id:3,name:"Night Star",cat:"Sleeper Coach",from:"Chennai",to:"Madurai",time:"22:00",price:700,seats:20,desc:"AC sleeper with
    blankets and water."}],
bookings:[],comments:[{bus:1,name:"Kala",text:"Comfortable journey and on time."}]}}
let S;try{S=JSON.parse(localStorage.getItem(K))}catch(e){}S=S||seed();
const save=()=>{try{localStorage.setItem(K,JSON.stringify(S))}catch(e){}};
let me=null,view="home",arg=null,q={from:"Chennai",to:"Bengaluru",date:""},pick=[];
const say=(m,bad)=>{$("msg").className=bad?"err":"ok";$("msg").textContent=m};
const taken=(b,d)=>S.bookings.filter(x=>x.bus==b&&x.date==d&&x.st=="Booked").flatMap(x=>x.seats);
const go=(v,a)=>{view=v;arg=a;pick=[];say("");render()};
const cities=()=>[...new Set(S.buses.flatMap(b=>[b.from,b.to]))];
function render(){
$("nav").innerHTML=`<button onclick="go('home')">Search Bus</button>`+(me?(me.admin?`<button
    onclick="go('acat')">Categories</button><button onclick="go('abus')">Buses</button><button
    onclick="go('ausers')">Users</button><button onclick="go('rep')">Reports</button>`:`<button onclick="go('my')">My
```

```
Bookings</button>`)+`<button onclick="go('pw')">Change Password</button><button class="s" onclick="me=null;go('home')">Logout (${esc(me.name)})</button>`:`<button onclick="go('login')">Login</button><button class="s" onclick="go('reg')">Register</button>`);
$("app").innerHTML=VIEW[view]()
const opts=(L,s)=>L.map(c=>`<option ${c==s?"selected":""}>${c}</option>`).join("");
const VIEW={
home:()=>`<h2>Search Buses</h2><div class="row"><div>From<select id="f">${opts(cities(),q.from)}</select></div><div>To<select id="t">${opts(cities(),q.to)}</select></div><div>Date<input id="d" type="date" min="${today()}" value="${q.date}"></div></div><button onclick="search()">Search</button>`+
(q.date?S.buses.filter(b=>b.from==q.from&&b.to==q.to).map(b=>`<div class="bus"><div><b>${esc(b.name)}</b> (${esc(b.cat)})<br>Departs ${b.time} | Rs.${b.price}/seat<br><span class="ok">${b.seats-taken(b.id,q.date).length} seats available</span></div><div><button onclick="go('detail',${b.id})">Details</button><button onclick="sel(${b.id})">Select Seats</button></div></div>`).join("")||"<p>No buses on this route.</p>":""),
detail:()=>{const b=S.buses.find(x=>x.id==arg);return`<h2>${esc(b.name)}</h2><p>${esc(b.cat)} | ${b.from} to ${b.to} | ${b.time} | Rs.${b.price}<br>${esc(b.desc)}</p><h3>Comments</h3>${S.comments.filter(c=>c.bus==b.id).map(c=>`<p><b>${esc(c.name)}:</b> ${esc(c.text)}</p>`).join("")||"<p>No comments yet.</p>"}<textarea id="cm" rows="2" placeholder="Write a comment"></textarea><button onclick="comment(${b.id})">Post Comment</button>`},
seat:()=>{const b=S.buses.find(x=>x.id==arg),t=taken(b.id,q.date);return`<h2>${esc(b.name)} - ${q.date}</h2><p>Choose seats (grey = booked)</p><div class="seats">${Array.from({length:b.seats},(_,i)=>i+1).map(n=>`<button class="${pick.includes(n)?"sel":""}" ${t.includes(n)?"disabled":""} onclick="tog(${n})">${n}</button>`).join("")}</div>
${pick.length?`<h3>Passenger Details</h3>${pick.map(n=>`<div class="row"><input id="pn${n}" placeholder="Passenger name (seat ${n})"><input id="pa${n}" type="number" min="1" placeholder="Age"></div>`).join("")}<p><b>Total: Rs.${pick.length*b.price}</b></p><input id="card" placeholder="Card number (demo payment gateway)" maxlength="16"><button onclick="book(${b.id})">Pay and Book</button>`:""}`},
login:()=>`<h2>Login</h2><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="login()">Login</button><button onclick="go('forgot')">Forgot Password</button><p>Demo: kala@gmail.com / 1234, admin@gobus.com / admin123</p>`,
forgot:()=>`<h2>Forgot Password</h2><input id="e" type="email" placeholder="Registered email"><button onclick="forgot()">Reset Password</button>`,
reg:()=>`<h2>Register</h2><input id="n" placeholder="Full name"><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="reg()">Create Account</button>`,
pw:()=>`<h2>Change Password</h2><input id="o" type="password" placeholder="Old password"><input id="p" type="password" placeholder="New password"><button onclick="chpw()">Update Password</button>`,
my:()=>`<h2>My Booking History</h2><div class="tw"><table><tr><th>ID</th><th>Bus</th><th>Date</th><th>Seats</th><th>Total</th><th>Status</th><th></th></tr>${S.bookings.filter(b=>b.uid==me.id).map(b=>`<tr><td>${b.id}</td><td>${esc(S.buses.find(x=>x.id==b.bus).name)}</td><td>${b.date}</td><td>${b.seats.join(", ")}</td><td>Rs.${b.total}</td><td>${b.st}</td><td>${b.st=="Booked"?`<button class="x" onclick="cancel(${b.id})">Cancel Ticket</button>`:"-"}</td></tr>`).join("")||`<tr><td colspan="7">No bookings yet.</td></tr>`}</table></div>`,
acat:()=>`<h2>Bus Categories</h2><p>${S.cats.map(esc).join(" | ")}</p><input id="c" placeholder="New category (Volvo, Sleeper coach...)"><button onclick="addCat()">Add Category</button>`,
abus:()=>{const b=S.buses.find(x=>x.id==arg)||{};return`<h2>${b.id?"Update":"Add"} Bus</h2><div class="row"><input id="bn" placeholder="Bus name" value="${esc(b.name)}"><select id="bc">${opts(S.cats,b.cat)}</select><input id="bf" placeholder="From" value="${esc(b.from)}"><input id="bt" placeholder="To" value="${esc(b.to)}"><input id="bh" type="time" value="${b.time||"20:00"}"><input id="bp" type="number" placeholder="Price" value="${b.price||""}"><input id="bs" type="number" placeholder="Seats (max 40)" value="${b.seats||20}"></div><textarea id="bd" rows="2" placeholder="Description">${esc(b.desc)}</textarea><button onclick="saveBus(${b.id||0})">${b.id?"Update":"Add"} Bus</button>
<div class="tw"><table><tr><th>Bus</th><th>Category</th><th>Route</th><th>Seats</th><th></th></tr>${S.buses.map(x=>`<tr><td>${esc(x.name)}</td><td>${esc(x.cat)}</td><td>${esc(x.from)} - ${esc(x.to)}</td><td>${x.seats}</td><td><button onclick="go('abus',${x.id})">Edit</button><button class="x" onclick="delBus(${x.id})">Delete</button></td></tr>`).join("")}</table></div>
<h3>Comments</h3>${S.comments.map((c,i)=>`<p>${esc(c.name)}: ${esc(c.text)} <button class="x" onclick="S.comments.splice(${i},1);save();render()">Delete</button></p>`).join("")}`},
ausers:()=>`<h2>Registered Users</h2><div class="tw"><table><tr><th>Name</th><th>Email</th><th></th></tr>${S.users.filter(u=>!u.admin).map(u=>`<tr><td>${esc(u.name)}</td><td>${esc(u.email)}</td><td><button class="x" onclick="S.users=S.users.filter(x=>x.id!=${u.id});save();render()">Remove</button></td></tr>`).join("")}</table></div>`,
rep:()=>{const ok=S.bookings.filter(b=>b.st=="Booked");return`<h2>Report</h2><p>Total tickets booked: <b>${ok.length}</b> | Cancelled: <b>${S.bookings.length-ok.length}</b> | Revenue: <b>Rs.${ok.reduce((s,b)=>s+b.total,0)}</b></p><div class="tw"><table><tr><th>Bus</th><th>Tickets</th><th>Seats sold</th><th>Revenue</th></tr>${S.buses.map(x=>{const L=ok.filter(b=>b.bus==x.id);return`<tr><td>${esc(x.name)}</td><td>${L.length}</td><td>${L.reduce((s,b)=>s+b.seats.length,0)}</td><td>Rs.${L.reduce((s,b)=>s+b.total,0)}</td></tr>`}).join("")}</table></div>`}};
function search(){q={from:V("f"),to:V("t"),date:V("d")};if(q.from==q.to||!q.date||q.date<today())return say("Choose different cities and a valid date.",1);render()}
function sel(id){if(!me){go("login");return say("Please login to book tickets.",1)}go("seat",id)}
```

```
function tog(n){const keep={};pick.forEach(x=>keep[x]=[$("pn"+x)?.value,$("pa"+x)?.value]);pick=pick.includes(n)?pick.filter(x=>x!=n):[...pick,n].sort((a,b)=>a-b);if(pick.length>6){pick.pop();say("Maximum 6 seats.",1)}render();pick.forEach(x=>{if(keep[x]){$("pn"+x).value=keep[x][0]||"";$("pa"+x).value=keep[x][1]||""}})}
function book(id){const b=S.buses.find(x=>x.id==id);if(pick.some(n=>!V("pn"+n)||!V("pa"+n)))return say("Enter name and age for every passenger.",1);if(V("card").length<12)return say("Enter a valid card number.",1);
if(pick.some(n=>taken(id,q.date).includes(n)))return say("A seat was just taken. Choose again.",1);
S.bookings.push({id:S.id++,uid:me.id,bus:id,date:q.date,seats:[...pick],pass:pick.map(n=>V("pn"+n)),total:pick.length*b.price,st:"Booked"});save();go("my");say("Payment successful. Confirmation email sent to "+me.email)}
function cancel(id){S.bookings.find(b=>b.id==id).st="Cancelled";save();render();say("Ticket cancelled.")}
function comment(id){if(!me)return say("Login to post a comment.",1);if(!V("cm"))return;S.comments.push({bus:id,name:me.name,text:V("cm")});save();render()}
function login(){const u=S.users.find(u=>u.email==V("e").toLowerCase()&&u.pw==$("p").value);if(!u)return say("Wrong email or password.",1);me=u;go(u.admin?"abus":"home");say("Welcome, "+u.name)}
function reg(){const e=V("e").toLowerCase();if(!V("n")||!e||$("p").value.length<4)return say("Enter name, email and a 4+ character password.",1);if(S.users.some(u=>u.email==e))return say("Email already registered.",1);const u={id:S.id++,name:V("n"),email:e,pw:$("p").value};S.users.push(u);save();me=u;go("home");say("Registered successfully.")}
function forgot(){const u=S.users.find(u=>u.email==V("e").toLowerCase());if(!u)return say("Email not found.",1);u.pw="1234";save();go("login");say("Password reset to 1234 (demo email sent). Please login.")}
function chpw(){if($("o").value!=me.pw)return say("Old password is wrong.",1);if($("p").value.length<4)return say("New password is too short.",1);me.pw=$("p").value;save();say("Password changed.")}
function addCat(){const c=V("c");if(!c||S.cats.includes(c))return say("Enter a new category name.",1);S.cats.push(c);save();render();say("Category added.")}
function saveBus(id){const o={name:V("bn"),cat:V("bc"),from:V("bf"),to:V("bt"),time:V("bh"),price:+V("bp"),seats:Math.min(40,+V("bs")||20),desc:V("bd")};if(!o.name||!o.from||!o.to||!o.price)return say("Fill name, route and price.",1);
if(id)Object.assign(S.buses.find(b=>b.id==id),o);else S.buses.push({id:S.id++,...o});save();go("abus");say("Bus saved.")}
function delBus(id){S.buses=S.buses.filter(b=>b.id!=id);save();render()}
render();
</script></body></html>
```

Output

The bus website lists buses for a route and date with available seats. The user picks seats on a seat map, enters passenger details and pays. The administrator adds categories, adds or updates buses, manages users and comments, and views the report.

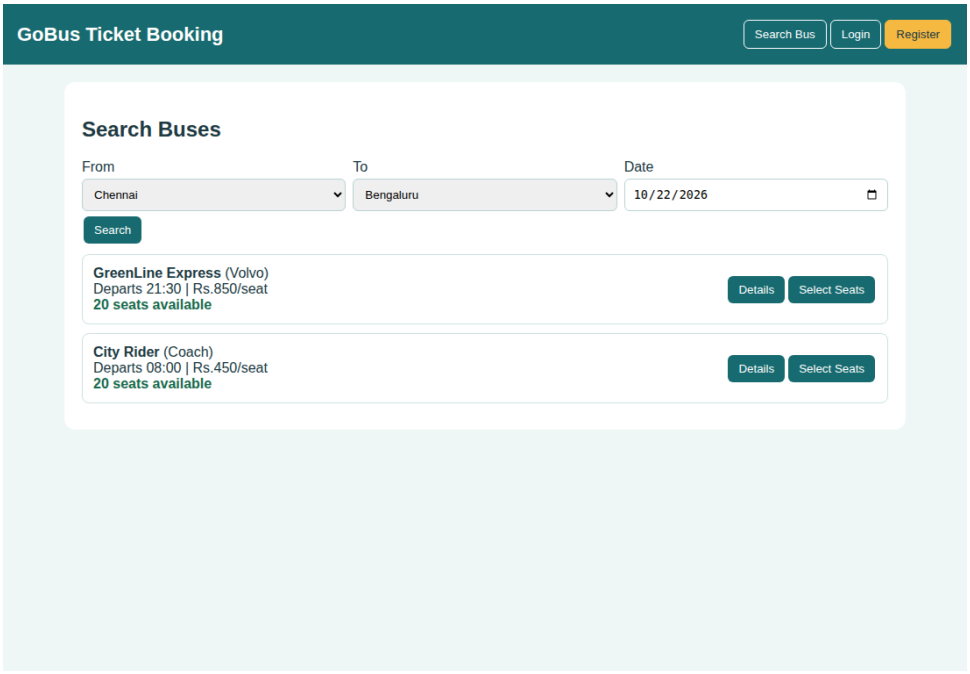


Figure 1: Bus search results

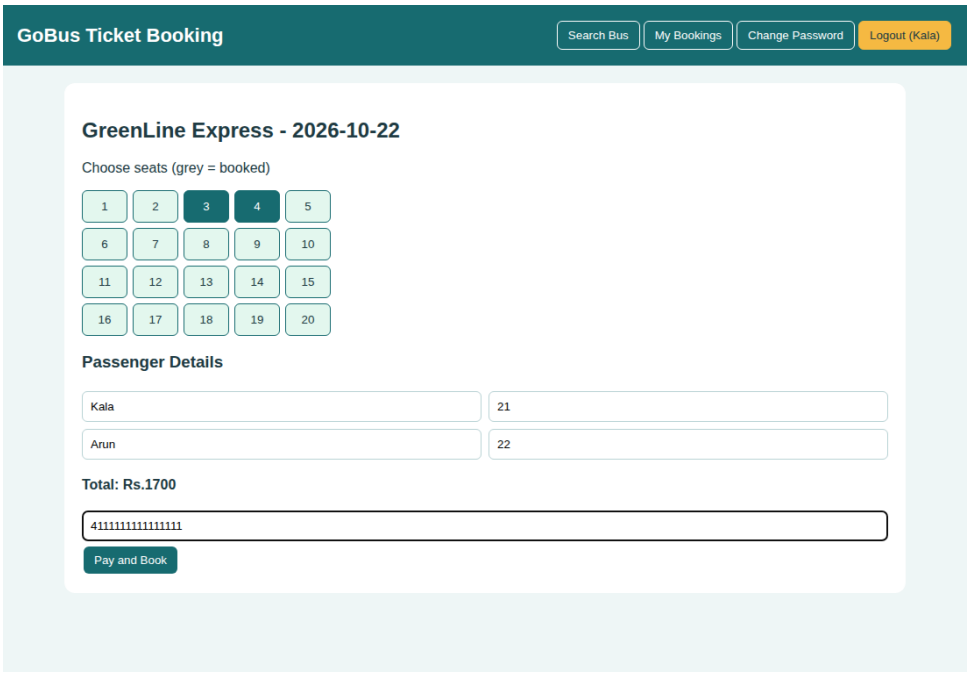


Figure 2: Seat selection and passenger details

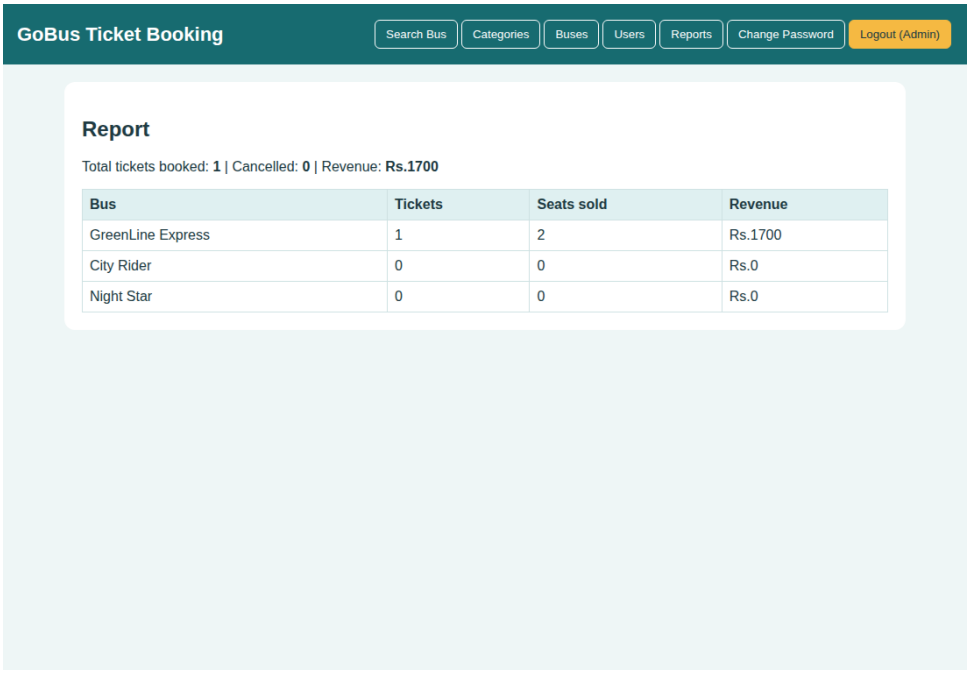


Figure 3: Administrator report

Result

Thus, an online bus ticket booking website with seat selection, cancellation, comments and report generation was created successfully.

Experiment 4: Online Train Reservation System

Aim

To create a train reservation website where a logged-in passenger searches trains, sees seat availability, books tickets with online or offline payment, receives a PNR, checks PNR status (CNF, RAC, WL), cancels with refund rules, and where the admin manages trains, fares, payments and refunds.

HTML Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>RailConnect Train Reservation</title>
<style>
body{font-family:Arial;background:#f0f4fa;margin:0;color:#243650}
header{background:#203f68;color:#fff;padding:16px;display:flex;flex-wrap:wrap;gap:8px;align-items:center}
header h1{font-size:22px;margin:0 auto 0 0}
button{background:#267b83;color:#fff;border:0;border-radius:6px;padding:8px 12px;cursor:pointer;margin:2px}
nav button{background:transparent;border:1px solid #fff}nav button.s{background:#f5b942;color:#203f68;border-color:#f5b942}
button.x{background:#b3261e}
main{max-width:940px;margin:20px auto;background:#fff;padding:20px;border-radius:12px}
input,select{padding:9px;border:1px solid #bccbe0;border-radius:6px;box-sizing:border-box;margin:4px 0;width:100%}
input[type=radio]{width:auto}
.tr{border:1px solid #d3deee;border-radius:8px;padding:12px;margin:10px 0}
.tw{overflow-x:auto}table{width:100%;border-collapse:collapse}th,td{border:1px solid
    #d3deee;padding:8px;text-align:left}th{background:#e4edf8}
.ok{color:#176b4d;font-weight:bold}.err{color:#b3261e;font-weight:bold}
.row{display:flex;gap:8px;flex-wrap:wrap}.row>*{flex:1;min-width:130px}
</style></head><body>
<header><h1>RailConnect</h1><nav id="nav"></nav></header>
<main id="app"></main><p id="msg" style="text-align:center"></p>
<script>
const
    $=i=>document.getElementById(i),V=i=>$(i).value.trim(),K="train_lab",CL=["SL","3A","2A"],CN={SL:"Sleeper",["3A"]:"AC
    3 Tier",["2A"]:"AC 2 Tier"};
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
const fmt=d=>d.getFullYear()+"-"+String(d.getMonth()+1).padStart(2,"0")+"-"+String(d.getDate()).padStart(2,"0");
const today=()=>fmt(new Date());
const days=(a,b)=>Math.round((new Date(b)-new Date(a))/864e5);
function seed(){return{id:10,
users:[{id:1,name:"Admin",email:"admin@rail.com",pw:"admin123",admin:1},{id:2,name:"Kala",email:"kala@gmail.com",pw:"12
    34"}],
trains:[{no:"12601",name:"Chennai Express",from:"Chennai",to:"Bengaluru",dep:"06:00",arr:"12:30",fare:{SL:450,"3A":1000,"2A"
    :1600},seats:{SL:4,"3A":3,"2A":2}},
{no:"12605",name:"Southern Superfast",from:"Chennai",to:"Bengaluru",dep:"08:15",arr:"15:00",fare:{SL:520,"3A":1100,"2A":170
    0},seats:{SL:4,"3A":3,"2A":2}},
{no:"12639",name:"Pandian
    Express",from:"Chennai",to:"Madurai",dep:"21:00",arr:"06:00",fare:{SL:600,"3A":1400,"2A":2000},seats:{SL:4,"3A":3,"2A":2}}],
tk:[]}}
let S;try{S=JSON.parse(localStorage.getItem(K))}catch(e){}S=S||seed();
const save=()=>{try{localStorage.setItem(K,JSON.stringify(S))}catch(e){}};
let me=null,view="home",arg=null,q={from:"Chennai",to:"Bengaluru",date:"",cls:"SL"},pnr="";
const say=(m,bad)=>{$("msg").className=bad?"err":"ok";$("msg").textContent=m};
const go=(v,a)=>{view=v;arg=a;say("");render()};
const
    used=(no,d,c,st)=>S.tk.filter(t=>t.train==no&&t.date==d&&t.cls==c&&t.st!="Cancelled"&&t.st!="Refunded"&&t.st!="Cancel
    Requested").flatMap(t=>t.pax).filter(p=>p.st==st).length;
const free=(t,d,c)=>t.seats[c]-used(t.no,d,c,"CNF");
const cities=()=>[...new Set(S.trains.flatMap(t=>[t.from,t.to]))];
```

```
const opts=(L,s)=>L.map(c=>`<option ${c==s?"selected":""}>${c}</option>`).join("");
const refundPct=(d)=>{const n=days(today(),d);return n>=7?90:n>=2?50:25};
function render(){
$("nav").innerHTML=`<button onclick="go('home')">Book Ticket</button><button onclick="go('pnr')">PNR Status</button>`+(me?(me.admin?`<button onclick="go('atrain')">Trains &amp;amp; Fares</button><button onclick="go('apay')">Payments &amp;amp; Refunds</button>`:`<button onclick="go('my')">My Tickets</button><button onclick="go('cancel')">Cancel Ticket</button>`)+`<button class="s" onclick="me=null;go('home')">Logout (${esc(me.name)})</button>`:`<button onclick="go('login')">Login</button><button class="s" onclick="go('reg')">Register</button>`);
$("app").innerHTML=VIEW[view]()}
const VIEW={
home:()=>`<h2>Journey Details</h2><div class="row"><div>From<select id="f">${opts(cities(),q.from)}</select></div><div>To<select id="t">${opts(cities(),q.to)}</select></div><div>Date<input id="d" type="date" min="${today()}" value="${q.date}"></div><div>Class<select id="c">${CL.map(c=>`<option value="${c}" ${c==q.cls?"selected":""}>${CN[c]}</option>`).join("")}</select></div></div><button onclick="search()">Search Trains</button>`+
(q.date?S.trains.filter(t=>t.from==q.from&&t.to==q.to).map(t=>`<div class="tr"><b>${esc(t.name)}</b> (${t.no}) | ${t.dep} - ${t.arr}<br>${CN[q.cls]}: Rs.${t.fare[q.cls]} per passenger | <span class="${free(t,q.date,q.cls)>0?"ok":"err"}">${free(t,q.date,q.cls)>0?free(t,q.date,q.cls)+" seats available":"Waiting list / RAC"}</span><br><button onclick="sel('${t.no}')">Book</button></div>`).join("")||"<p>No trains on this route.</p>":""),
book:()=>{const t=S.trains.find(x=>x.no==arg);return`<h2>${esc(t.name)} - ${q.date}</h2><p>${t.from} to ${t.to} | ${CN[q.cls]} | Rs.${t.fare[q.cls]} per passenger</p><label>Number of passengers</label><select id="n" onchange="paxRows()">${[1,2,3,4,5,6].map(i=>`<option>${i}</option>`).join("")}</select><div id="pax"></div><p>Payment mode</p><label><input type="radio" name="pm" value="Online" checked> Online</label> <label><input type="radio" name="pm" value="Offline"> Offline (pay at counter)</label><input id="card" placeholder="Card number (for online payment)" maxlength="16"><button onclick="book('${t.no}')">Pay and Generate Ticket</button>`},
login:()=>`<h2>Login</h2><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="login()">Login</button><p>Demo: kala@gmail.com / 1234, admin@rail.com / admin123</p>`,
reg:()=>`<h2>Register</h2><input id="n" placeholder="Full name"><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="reg()">Create Account</button>`,
pnr:()=>`<h2>Check PNR Status</h2><input id="pn" placeholder="Enter 10-digit PNR" value="${pnr}"><button onclick="pnr=V('pn');render();$('pn').value=pnr">Check Status</button>`+(()=>{const t=S.tk.find(x=>x.pnr==pnr);return pnr?t?tkt(t):`<p class="err">PNR not found.</p>`:""})(),
my:()=>`<h2>My Tickets</h2>${S.tk.filter(t=>t.uid==me.id).map(tkt).join("")||"<p>No tickets yet.</p>"}`,
cancel:()=>`<h2>Cancel Ticket</h2><p>Refund rules: 7+ days before journey 90%, 2 to 6 days 50%, less than 2 days 25%.</p><input id="pn" placeholder="Enter PNR number"><button class="x" onclick="cancelReq()">Send Cancel Request</button>`,
atrain:()=>`<h2>Manage Trains and Fares</h2><div class="tw"><table><tr><th>No</th><th>Train</th><th>Route</th><th>Fare SL / 3A / 2A</th><th>Seats SL / 3A / 2A</th><th></th></tr>${S.trains.map((t,i)=>`<tr><td>${t.no}</td><td>${esc(t.name)}</td><td>${t.from} - ${t.to}</td><td>${CL.map(c=>`<input style="width:70px" id="f${i}${c}" type="number" value="${t.fare[c]}">`).join("")}</td><td>${CL.map(c=>`<input style="width:60px" id="s${i}${c}" type="number" value="${t.seats[c]}">`).join("")}</td><td><button onclick="upd(${i})">Update</button></td></tr>`).join("")}</table></div>
<h3>Add Train</h3><div class="row"><input id="an" placeholder="Train no"><input id="am" placeholder="Name"><input id="af" placeholder="From"><input id="at" placeholder="To"><input id="ad" type="time" value="06:00"><input id="aa" type="time" value="12:00"><input id="ap" type="number" placeholder="Sleeper fare"></div><button onclick="addTrain()">Add Train</button>`,
apay:()=>`<h2>Payments and Refunds</h2><div class="tw"><table><tr><th>PNR</th><th>Passenger</th><th>Mode</th><th>Paid</th><th>Status</th><th></th></tr>${S.tk.map(t=>`<tr><td>${t.pnr}</td><td>${esc(t.pax[0].n)}${t.pax.length>1?" +"+(t.pax.length-1):""}</td><td>${t.mode}</td><td>Rs.${t.total}</td><td>${t.st}${t.refund?" (refund Rs."+t.refund+")":""}</td><td>${t.st=="Cancel Requested"?`<button onclick="refund('${t.pnr}')">Pass Refund</button>`:""}</td></tr>`).join("")||`<tr><td colspan="6">No payments yet.</td></tr>`}</table></div>`};
const tkt=t=>`<div class="tr"><b>PNR ${t.pnr}</b> | ${esc(S.trains.find(x=>x.no==t.train).name)} | ${t.date} | ${CN[t.cls]}<br>Fare paid: Rs.${t.total} (${t.mode}) | Ticket: <b>${t.st}</b>${t.refund?" | Refund: Rs."+t.refund:""}<br>${t.pax.map(p=>`${esc(p.n)} (${p.age}) - <span class="${p.st=="CNF"?"ok":"err"}">${p.st}</span>`).join("<br>")}</div>`;
function search(){q={from:V("f"),to:V("t"),date:V("d"),cls:V("c")};if(q.from==q.to||!q.date||q.date<today())return say("Choose different stations and a valid date.",1);render()}
function sel(no){if(!me){go("login");return say("Please login with valid credentials to reserve tickets.",1)}go("book",no);paxRows()}
function paxRows(){$("pax").innerHTML=Array.from({length:+V("n")},(_,i)=>`<div class="row"><input id="pn${i}" placeholder="Passenger ${i+1} name"><input id="pa${i}" type="number" min="1" placeholder="Age"></div>`).join("")}
function book(no){const t=S.trains.find(x=>x.no==no),n=+V("n"),mode=document.querySelector("[name=pm]:checked").value;
for(let i=0;i<n;i++)if(!V("pn"+i)||!V("pa"+i))return say("Enter name and age for every passenger.",1);
if(mode=="Online"&&V("card").length<12)return say("Enter a valid card number for online payment.",1);
```

```
let cnf=free(t,q.date,q.cls),rac=2-used(no,q.date,q.cls,"RAC");
const pax=Array.from({length:n},(_,i)=>{const st=cnf>0?(cnf--,"CNF"):rac>0?(rac--,"RAC"):"WL";return{n:V("pn"+i),age:V("pa"+i),st}});
const p="9"+String(Math.floor(Math.random()*1e9)).padStart(9,"0");
S.tk.push({pnr:p,uid:me.id,train:no,date:q.date,cls:q.cls,pax,total:n*t.fare[q.cls],mode,st:"Booked"});save();pnr=p;go("my");say("Payment recorded. Ticket generated. Your PNR is "+p)}
function cancelReq(){const t=S.tk.find(x=>x.pnr==V("pn")&&x.uid==me.id);if(!t)return say("PNR not found in your account.",1);if(t.st!="Booked")return say("Ticket is already "+t.st+".",1);
t.refund=Math.round(t.total*refundPct(t.date)/100);t.st="Cancel Requested";save();say("Cancel request sent. Refund of Rs."+t.refund+" will be passed by the admin.")}
function refund(p){const t=S.tk.find(x=>x.pnr==p);t.st="Refunded";save();render();say("Refund of Rs."+t.refund+" passed to passenger.")}
function login(){const u=S.users.find(u=>u.email==V("e").toLowerCase()&&u.pw==$("p").value);if(!u)return say("Wrong email or password.",1);me=u;go(u.admin?"atrain":"home");say("Welcome, "+u.name)}
function reg(){const e=V("e").toLowerCase();if(!V("n")||!e||$("p").value.length<4)return say("Enter name, email and a 4+ character password.",1);if(S.users.some(u=>u.email==e))return say("Email already registered.",1);const u={id:S.id++,name:V("n"),email:e,pw:$("p").value};S.users.push(u);save();me=u;go("home");say("Registered successfully.")}
function upd(i){CL.forEach(c=>{S.trains[i].fare[c]=+V("f"+i+c);S.trains[i].seats[c]=+V("s"+i+c)});save();say("Train updated.")}
function addTrain(){if(!V("an")||!V("am")||!V("af")||!V("at")||!+V("ap"))return say("Fill all train details.",1);const p=+V("ap");S.trains.push({no:V("an"),name:V("am"),from:V("af"),to:V("at"),dep:V("ad"),arr:V("aa"),fare:{SL:p,"3A":Math.round(p*2.2),"2A":Math.round(p*3.4)},seats:{SL:4,"3A":3,"2A":2}});save();render();say("Train added.")}
render();
</script></body></html>
```

Output

The train website lists trains with class-wise availability. The passenger books, pays and receives a PNR with CNF, RAC or WL status. Cancelling by PNR calculates the refund by the refund rules, and the administrator passes the refund amount.

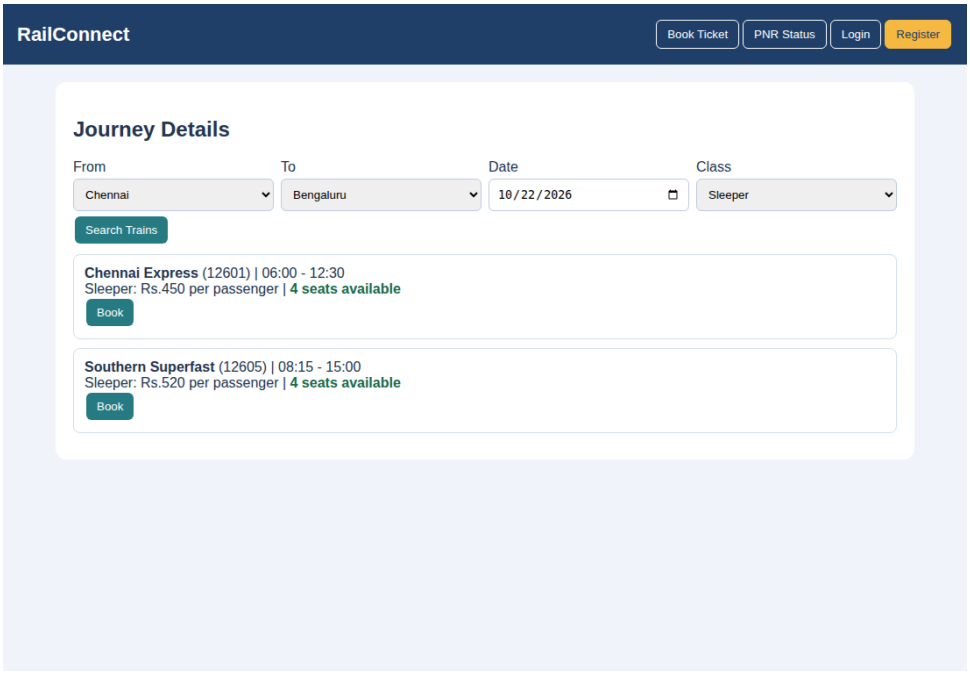


Figure 1: Train search with availability

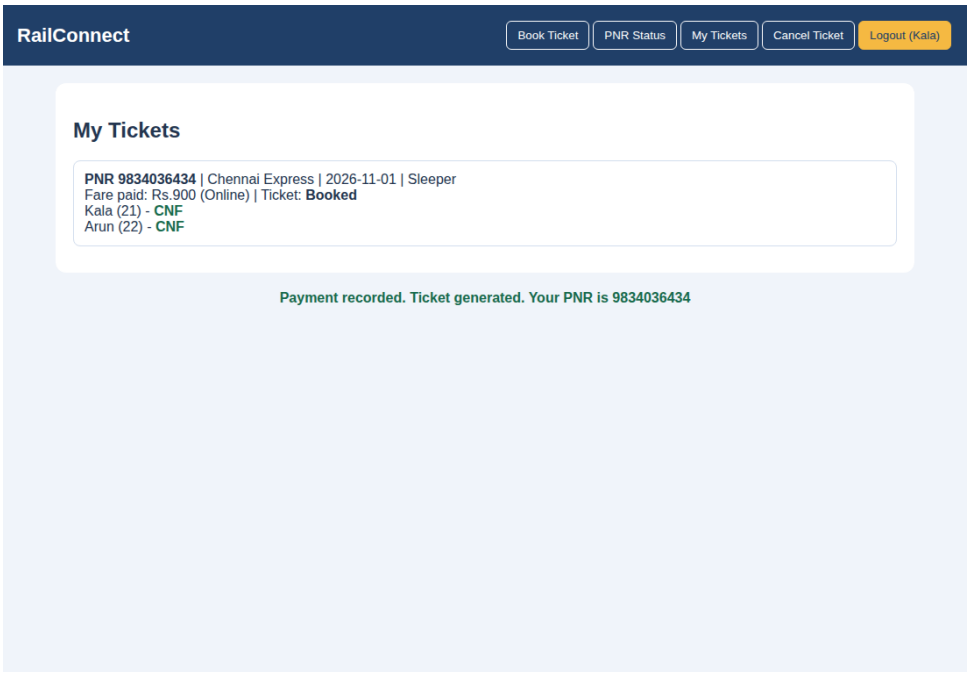


Figure 2: Ticket with PNR and status

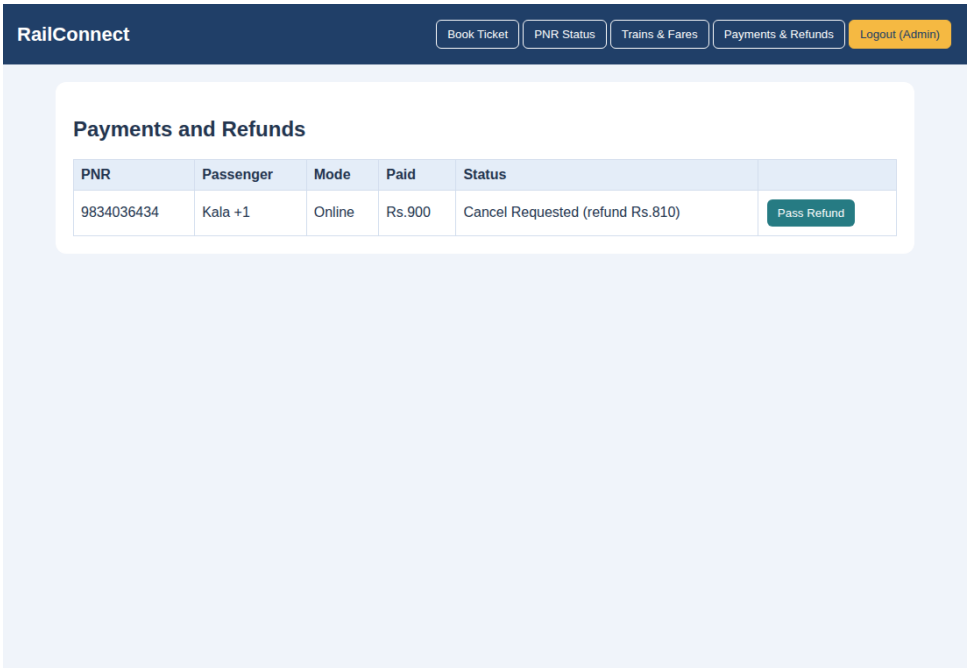


Figure 3: Admin: payments and refunds

Result

Thus, an online train reservation website with PNR status, cancellation and refund handling was created successfully.

Experiment 5: Online Flight Reservation System

Aim

To develop a flight reservation website with flight search by route, date and class, seat selection, passenger details, payment, e-ticket with PNR, ticket printing and cancellation, forgot password, customer feedback, and an admin panel for flights and reports.

HTML Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>SkyNova Flight Reservation</title>
<style>
body{font-family:Arial;background:#f1f6fb;margin:0;color:#20344f}
header{background:#142b49;color:#fff;padding:16px;display:flex;flex-wrap:wrap;gap:8px;align-items:center}
header h1{font-size:22px;margin:0 auto 0 0}
button{background:#168c91;color:#fff;border:0;border-radius:6px;padding:8px 12px;cursor:pointer;margin:2px}
nav button{background:transparent;border:1px solid #fff}nav button.s{background:#5fd6c8;color:#142b49;border-color:#5fd6c8}
button.x{background:#b3261e}
main{max-width:940px;margin:20px auto;background:#fff;padding:20px;border-radius:12px}
input,select,textarea{padding:9px;border:1px solid #b9c9de;border-radius:6px;box-sizing:border-box;margin:4px 0;width:100%}
.fl{border:1px solid #d3deee;border-radius:8px;padding:12px;margin:10px 0}
.seats{display:grid;grid-template-columns:repeat(6,48px);gap:6px;margin:10px 0}
.seats button{margin:0;padding:9px 0;background:#e3f7f4;color:#20344f;border:1px solid #168c91}
.seats button.sel{background:#168c91;color:#fff}.seats button:disabled{background:#ccc;border-color:#aaa;color:#777}
.tw{overflow-x:auto}table{width:100%;border-collapse:collapse}th,td{border:1px solid
    #d3deee;padding:8px;text-align:left}th{background:#e4edf8}
.ok{color:#176b4d;font-weight:bold}.err{color:#b3261e;font-weight:bold}
.row{display:flex;gap:8px;flex-wrap:wrap}.row>*{flex:1;min-width:130px}
@media print{header,#msg,button{display:none}body{background:#fff}main{box-shadow:none}}
</style></head><body>
<header><h1>SkyNova Flight Booking</h1><nav id="nav"></nav></header>
<main id="app"></main><p id="msg" style="text-align:center"></p>
<script>
const $=i=>document.getElementById(i),V=i=>$(i).value.trim(),K="flight_lab";
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
const fmt=d=>d.getFullYear()+"-"+String(d.getMonth()+1).padStart(2,"0")+"-"+String(d.getDate()).padStart(2,"0");
const today=()=>fmt(new Date());
function seed(){return{id:10,
users:[{id:1,name:"Admin",email:"admin@sky.com",pw:"admin123",admin:1},{id:2,name:"Kala",email:"kala@gmail.com",pw:"12
    34"}],
flights:[{no:"SN204",air:"SkyNova",from:"Chennai",to:"Mumbai",dep:"07:30",arr:"09:40",price:4200,seats:12,biz:6},
{no:"SN310",air:"SkyNova",from:"Chennai",to:"Delhi",dep:"10:15",arr:"13:00",price:5600,seats:12,biz:6},
{no:"AI441",air:"Air India",from:"Mumbai",to:"Delhi",dep:"18:00",arr:"20:10",price:4800,seats:12,biz:6},
{no:"IG512",air:"IndiGo",from:"Chennai",to:"Mumbai",dep:"16:45",arr:"18:55",price:3900,seats:12,biz:6}],
bk:[],fb:[]}}
let S;try{S=JSON.parse(localStorage.getItem(K))}catch(e){}S=S||seed();
const save=()=>{try{localStorage.setItem(K,JSON.stringify(S))}catch(e){}};
let me=null,view="home",arg=null,q={from:"Chennai",to:"Mumbai",date:"",cls:"Economy"},pick=null;
const MULT={Economy:1,Business:2.5};
const say=(m,bad)=>{$("msg").className=bad?"err":"ok";$("msg").textContent=m};
const go=(v,a)=>{view=v;arg=a;pick=null;say("");render()};
const taken=(no,d,c)=>S.bk.filter(b=>b.fl==no&&b.date==d&&b.cls==c&&b.st=="Confirmed").map(b=>b.seat);
const cap=(f,c)=>c=="Business"?f.biz:f.seats;
const cities=()=>[...new Set(S.flights.flatMap(f=>[f.from,f.to]))];
const opts=(L,s)=>L.map(c=>`<option ${c==s?"selected":""}>${c}</option>`).join("");
const fare=(f,c)=>Math.round(f.price*MULT[c]);
function render(){
```

```
$("nav").innerHTML=`<button onclick="go('home')">Search Flight</button><button onclick="go('fb')">Feedback</button>`+(me?(me.admin?`<button onclick="go('aflt')">Flights</button><button onclick="go('abk')">Bookings</button><button onclick="go('rep')">Report</button>`:`<button onclick="go('my')">My Tickets</button>`)+`<button class="s" onclick="me=null;go('home')">Logout (${esc(me.name)})</button>`:`<button onclick="go('login')">Login</button><button class="s" onclick="go('reg')">Register</button>`);
$("app").innerHTML=VIEW[view]()
const seatName=n=>Math.ceil(n/6)+"ABCDEF"[(n-1)%6];
const VIEW={
home:()=>`<h2>Search Flights</h2><div class="row"><div>From<select id="f">${opts(cities(),q.from)}</select></div><div>To<select id="t">${opts(cities(),q.to)}</select></div><div>Date<input id="d" type="date" min="${today()}" value="${q.date}"></div><div>Class<select id="c">${opts(["Economy","Business"],q.cls)}</select></div></div><button onclick="search()">Search Flights</button>`+
(q.date?S.flights.filter(f=>f.from==q.from&&f.to==q.to).map(f=>{const left=cap(f,q.cls)-taken(f.no,q.date,q.cls).length;return`<div class="fl"><b>${f.air} ${f.no}</b> | ${f.dep} - ${f.arr}<br>${q.cls} fare: Rs.${fare(f,q.cls)} | <span class="${left>0?"ok":"err"}">${left} seats left</span><br><button ${left>0?"":"disabled"} onclick="sel('${f.no}')">Select Flight</button></div>`}).join("")||"<p>No flights on this route.</p>":""),
seat:()=>{const f=S.flights.find(x=>x.no==arg),t=taken(f.no,q.date,q.cls);return`<h2>${f.air} ${f.no} - ${q.date}</h2><p>Choose a ${q.cls} seat (grey = taken)</p><div class="seats">${Array.from({length:cap(f,q.cls)},(_,i)=>i+1).map(n=>`<button class="${pick==n?"sel":""}" ${t.includes(n)?"disabled":""} onclick="pick=${n};render()">${seatName(n)}</button>`).join("")}</div>${pick?`<h3>Passenger Details (seat ${seatName(pick)})</h3><div class="row"><input id="pn" placeholder="Passenger name"><input id="pa" type="number" min="1" placeholder="Age"></div><p><b>Total: Rs.${fare(f,q.cls)}</b></p><input id="card" placeholder="Card number (demo payment gateway)" maxlength="16"><button onclick="book('${f.no}')">Pay and Book Ticket</button>`:""}`},
login:()=>`<h2>Login</h2><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="login()">Login</button><button onclick="go('forgot')">Forgot Password</button><p>Demo: kala@gmail.com / 1234, admin@sky.com / admin123</p>`,
forgot:()=>`<h2>Forgot Password</h2><input id="e" type="email" placeholder="Registered email"><button onclick="forgot()">Reset Password</button>`,
reg:()=>`<h2>Register</h2><input id="n" placeholder="Full name"><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="reg()">Create Account</button>`,
my:()=>`<h2>My Tickets</h2>${S.bk.filter(b=>b.uid==me.id).map(b=>tkt(b,1)).join("")||"<p>No tickets yet.</p>"}`,
fb:()=>`<h2>Customer Feedback</h2><input id="fn" placeholder="Your name" value="${me?esc(me.name):""}"><select id="fr">${[5,4,3,2,1].map(i=>`<option>${i}</option>`).join("")}</select><textarea id="fm" rows="3" placeholder="Your feedback"></textarea><button onclick="feed()">Submit Feedback</button>`,
aflt:()=>{const f=S.flights.find(x=>x.no==arg)||{};return`<h2>${f.no?"Update":"Add"} Flight</h2><div class="row"><input id="an" placeholder="Flight no" value="${f.no||""}"><input id="aa" placeholder="Airline" value="${esc(f.air)}"><input id="af" placeholder="From" value="${esc(f.from)}"><input id="at" placeholder="To" value="${esc(f.to)}"><input id="ad" type="time" value="${f.dep||"08:00"}"><input id="ar" type="time" value="${f.arr||"10:00"}"><input id="ap" type="number" placeholder="Economy fare" value="${f.price||""}"><input id="as" type="number" placeholder="Economy seats" value="${f.seats||12}"><input id="ab" type="number" placeholder="Business seats" value="${f.biz||6}"></div><button onclick="saveFl()">${f.no?"Update":"Add"} Flight</button><div class="tw"><table><tr><th>Flight</th><th>Route</th><th>Time</th><th>Fare</th><th>Seats E/B</th><th></th></tr>${S.flights.map(x=>`<tr><td>${x.air} ${x.no}</td><td>${x.from} - ${x.to}</td><td>${x.dep}-${x.arr}</td><td>Rs.${x.price}</td><td>${x.seats}/${x.biz}</td><td><button onclick="go('aflt','${x.no}')">Edit</button><button class="x" onclick="S.flights=S.flights.filter(y=>y.no!='${x.no}');save();render()">Delete</button></td></tr>`).join("")}</table></div>`},
abk:()=>`<h2>All Bookings</h2>${S.bk.map(b=>tkt(b,0)).join("")||"<p>No bookings yet.</p>"}`,
rep:()=>{const ok=S.bk.filter(b=>b.st=="Confirmed");return`<h2>Admin Report</h2><p>Confirmed tickets: <b>${ok.length}</b> | Cancelled: <b>${S.bk.length-ok.length}</b> | Revenue: <b>Rs.${ok.reduce((s,b)=>s+b.total,0)}</b> | Users: <b>${S.users.filter(u=>!u.admin).length}</b></p><h3>Customer Feedback</h3>${S.fb.map(f=>`<p>${"*".repeat(f.r)} <b>${esc(f.n)}:</b> ${esc(f.m)}</p>`).join("")||"<p>No feedback yet.</p>"}`}};
const tkt=(b,u)=>{const f=S.flights.find(x=>x.no==b.fl)||{};return`<div class="fl"><b>E-Ticket PNR ${b.pnr}</b> | ${f.air} ${b.fl} | ${f.from} to ${f.to} | ${b.date} ${f.dep||""}<br>Passenger: ${esc(b.pn)} (${b.age}) | ${b.cls} seat ${seatName(b.seat)} | Rs.${b.total} | <b class="${b.st=="Confirmed"?"ok":"err"}">${b.st}</b><br>${u&&b.st=="Confirmed"?`<button onclick="print()">Print Ticket</button><button class="x" onclick="cancel('${b.pnr}')">Cancel Ticket</button>`:""}</div>`};
function search(){q={from:V("f"),to:V("t"),date:V("d"),cls:V("c")};if(q.from==q.to||!q.date||q.date<today())return say("Choose different cities and a valid date.",1);render()}
function sel(no){if(!me){go("login");return say("Please login to book a flight.",1)}go("seat",no)}
function book(no){const f=S.flights.find(x=>x.no==no);if(!V("pn")||!V("pa"))return say("Enter passenger name and age.",1);if(V("card").length<12)return say("Enter a valid card number.",1);
if(taken(no,q.date,q.cls).includes(pick))return say("Seat just taken. Choose another.",1);
S.bk.push({pnr:"SN"+Math.floor(1e5+Math.random()*9e5),uid:me.id,fl:no,date:q.date,cls:q.cls,seat:pick,pn:V("pn"),age:V("pa"),total:fare(f,q.cls),st:"Confirmed"});save();go("my");say("Payment successful. E-ticket sent to "+me.email)}
function cancel(p){S.bk.find(b=>b.pnr==p).st="Cancelled";save();render();say("Ticket cancelled.")}
```

```
function feed(){if(!V("fn")||!V("fm"))return say("Enter your name and feedback.",1);S.fb.push({n:V("fn"),r:+V("fr"),m:V("fm")});save();go("home");say("Thank you for your feedback.")}
function login(){const u=S.users.find(u=>u.email==V("e").toLowerCase()&&u.pw==$("p").value);if(!u)return say("Wrong email or password.",1);me=u;go(u.admin?"aflt":"home");say("Welcome, "+u.name)}
function reg(){const e=V("e").toLowerCase();if(!V("n")||!e||$("p").value.length<4)return say("Enter name, email and a 4+ character password.",1);if(S.users.some(u=>u.email==e))return say("Email already registered.",1);const u={id:S.id++,name:V("n"),email:e,pw:$("p").value};S.users.push(u);save();me=u;go("home");say("Registered successfully.")}
function forgot(){const u=S.users.find(u=>u.email==V("e").toLowerCase());if(!u)return say("Email not found.",1);u.pw="1234";save();go("login");say("Password reset to 1234 (demo email sent).")}
function saveFl(){const o={no:V("an"),air:V("aa"),from:V("af"),to:V("at"),dep:V("ad"),arr:V("ar"),price:+V("ap"),seats:+V("as")||12,biz:+V("ab")||6};if(!o.no||!o.from||!o.to||!o.price)return say("Fill flight no, route and fare.",1);
const i=S.flights.findIndex(f=>f.no==o.no);if(i>=0)S.flights[i]=o;else S.flights.push(o);save();go("aflt");say("Flight saved.")}
render();
</script></body></html>
```

Output

The flight website lists flights with seats left. The user selects a seat, pays and gets an e-ticket that can be printed or cancelled. The administrator adds or updates flights and views the report with feedback.

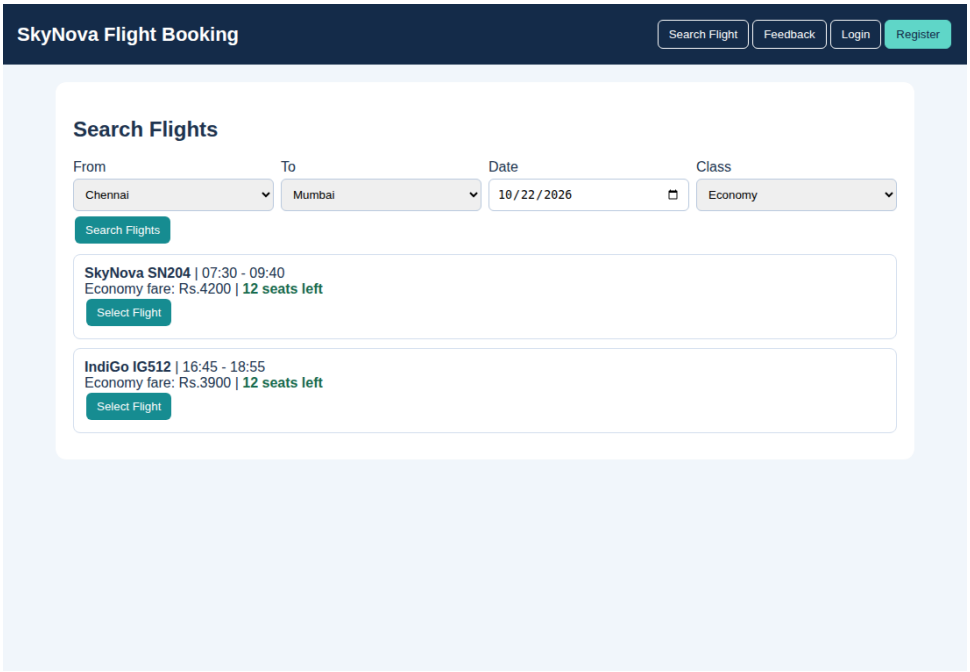


Figure 1: Flight search results

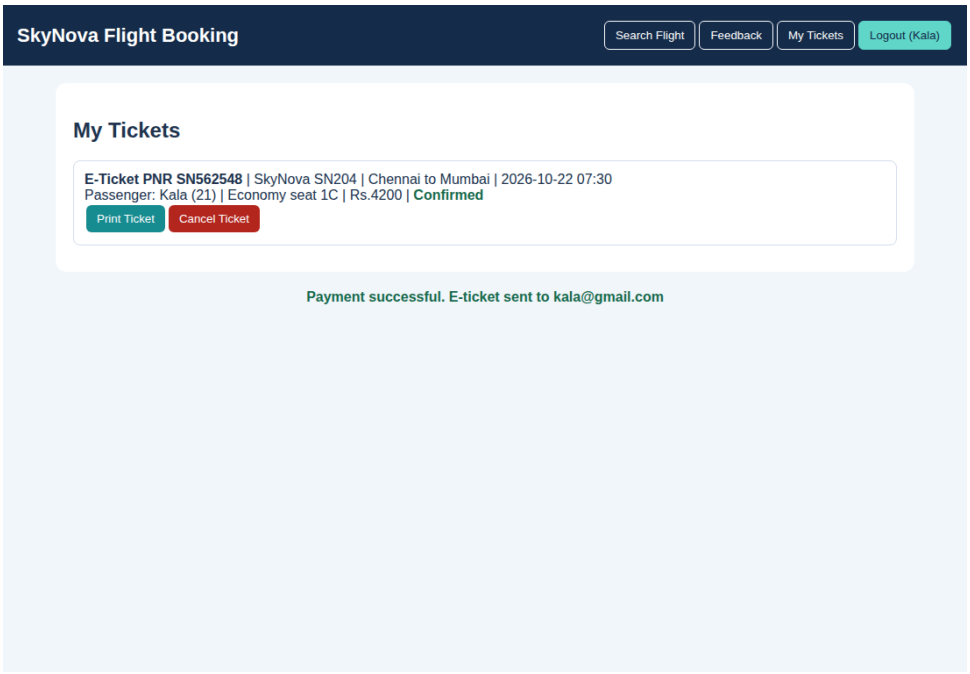


Figure 2: My ticket page

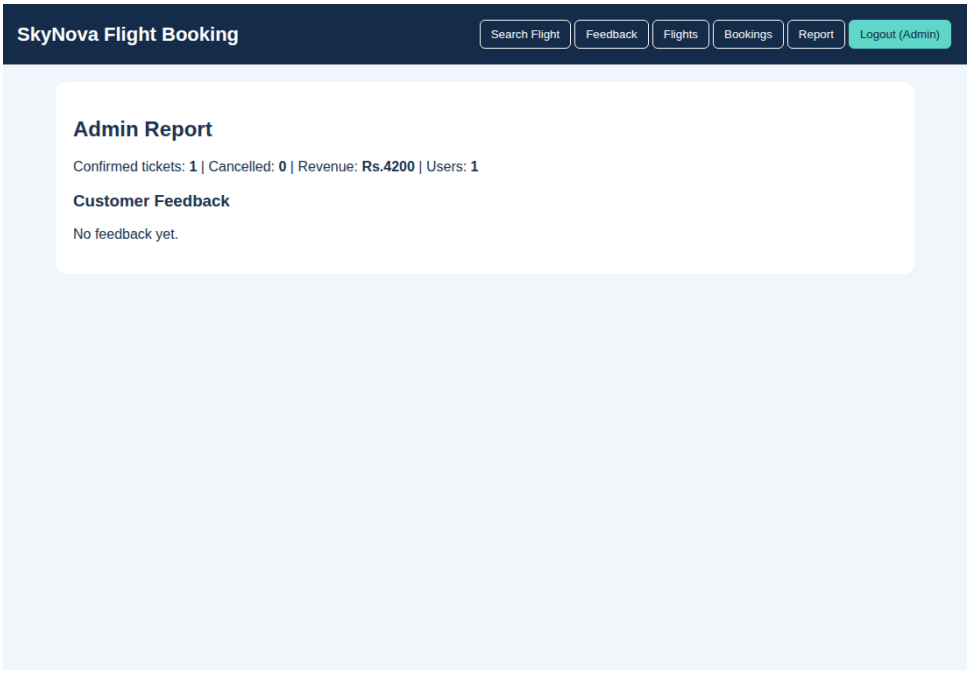


Figure 3: Admin report and feedback

Result

Thus, an online flight reservation website with seat booking, e-ticket, cancellation, feedback and admin report was created successfully.

Experiment 6: Online Cake Ordering System

Aim

To develop an online cake ordering website where students register, log in, search and filter cakes by category, sub category and price range, view descriptions, add to cart, place orders with a shipping address, and track shipping status, while the admin manages cakes and order status.

HTML Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Sweet Crumbs Cake Shop</title>
<style>
body{font-family:Arial;background:#fff6f7;margin:0;color:#573943}
header{background:#a84667;color:#fff;padding:16px;display:flex;flex-wrap:wrap;gap:8px;align-items:center}
header h1{font-size:22px;margin:0 auto 0 0}
button{background:#a84667;color:#fff;border:0;border-radius:6px;padding:8px 12px;cursor:pointer;margin:2px}
nav button{background:transparent;border:1px solid #fff}nav button.s{background:#ffd9e2;color:#a84667;border-color:#ffd9e2}
button.x{background:#b3261e}
main{max-width:960px;margin:20px auto;background:#fff;padding:20px;border-radius:12px}
input,select,textarea{padding:9px;border:1px solid #e6c3cd;border-radius:6px;box-sizing:border-box;margin:4px 0;width:100%}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(200px,1fr));gap:12px;margin-top:12px}
.cake{background:#fff0f3;border-radius:10px;padding:14px}
.cake .pic{height:70px;border-radius:8px;background:linear-gradient(135deg,#f7b6c6,#ffe3c4);display:flex;align-items:center;justify
-content:center;font-weight:bold;color:#a84667;margin-bottom:8px}
.tw{overflow-x:auto}table{width:100%;border-collapse:collapse}th,td{border:1px solid
#f0d3da;padding:8px;text-align:left}th{background:#ffe3ea}
.ok{color:#176b4d;font-weight:bold}.err{color:#b3261e;font-weight:bold}
.row{display:flex;gap:8px;flex-wrap:wrap}.row>*{flex:1;min-width:120px}
</style></head><body>
<header><h1>Sweet Crumbs</h1><nav id="nav"></nav></header>
<main id="app"></main><p id="msg" style="text-align:center"></p>
<script>
const $=i=>document.getElementById(i),V=i=>$(i).value.trim(),K="cake_lab";
const esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","":"&quot;","'":"&#39;"}[c]));
const STAGES=["Order Placed","Baking","Out for Delivery","Delivered"],SIZE={"500 g":1,"1 kg":1.8,"2 kg":3.4};
function seed(){return{id:20,
users:[{id:1,name:"Admin",email:"admin@sweetcrumbs.com",pw:"admin123",admin:1},{id:2,name:"Kala",email:"kala@gmail.com
",pw:"1234",addr:["12 Beach Road, Chennai - 600001"]}],
cakes:[{id:1,name:"Chocolate Dream",cat:"Birthday",sub:"Chocolate",price:650,stock:10,desc:"Rich chocolate sponge with creamy
ganache."},
{id:2,name:"Strawberry Delight",cat:"Birthday",sub:"Fruit",price:750,stock:8,desc:"Soft vanilla cake with fresh strawberry cream."},
{id:3,name:"Classic Vanilla",cat:"Anniversary",sub:"Vanilla",price:500,stock:12,desc:"Light, fluffy vanilla cream celebration
cake."},
{id:4,name:"Black Forest",cat:"Anniversary",sub:"Chocolate",price:850,stock:6,desc:"Chocolate layers, cherries and whipped
cream."},
{id:5,name:"Three Tier Wedding Cake",cat:"Wedding",sub:"Vanilla",price:3200,stock:3,desc:"Elegant fondant cake for
weddings."},
{id:6,name:"Pineapple Cream",cat:"Birthday",sub:"Fruit",price:600,stock:9,desc:"Pineapple chunks in soft cream sponge."}],
cart:[],orders:[]}}
let S;try{S=JSON.parse(localStorage.getItem(K))}catch(e){}S=S||seed();
const save=()=>{try{localStorage.setItem(K,JSON.stringify(S))}catch(e){}};
let me=null,view="home",arg=null,fl={s:"",cat:"",sub:"",min:"",max:""};
const say=(m,bad)=>{$("msg").className=bad?"err":"ok";$("msg").textContent=m};
const go=(v,a)=>{view=v;arg=a;say("");render()};
const cart=()=>S.cart.filter(c=>c.uid==(me&&me.id));
const price=c=>Math.round(S.cakes.find(k=>k.id==c.cake).price*SIZE[c.size])*c.qty;
const opts=(L,s,all)=>(all?`<option value="">${all}</option>`:"")+L.map(c=>`<option
${c==s?"selected":""}>${c}</option>`).join("");
```

```
function render(){
$("nav").innerHTML=`<button onclick="go('home')">Cakes</button>`+(me?(me.admin?`<button onclick="go('acake')">Manage Cakes</button><button onclick="go('aord')">Orders</button>`:`<button onclick="go('cart')">Cart (${cart().length})</button><button onclick="go('my')">My Orders</button>`)+`<button class="s" onclick="me=null;go('home')">Logout (${esc(me.name)})</button>`:`<button onclick="go('login')">Login</button><button class="s" onclick="go('reg')">Register</button>`);
$("app").innerHTML=VIEW[view]()}
function shown(){return S.cakes.filter(c=>(!fl.cat||c.cat==fl.cat)&&(!fl.sub||c.sub==fl.sub)&&(!fl.s||(c.name+c.desc).toLowerCase().includes(fl.s.toLowerCase()))&&(!fl.min||c.price>=+fl.min)&&(!fl.max||c.price<=+fl.max))}
function cards(){return shown().map(c=>`<div class="cake"><div class="pic">${esc(c.sub)}</div><b>${esc(c.name)}</b><br><small>${esc(c.cat)} | ${esc(c.sub)}</small><br>Rs.${c.price} / 500 g<br><span class="${c.stock?"ok":"err"}">${c.stock?c.stock+" in stock":"Out of stock"}</span><br><button onclick="go('view',${c.id})">View</button></div>`).join("")||"<p>No cakes match your filters.</p>"}
function filt(){fl={s:V("s"),cat:V("c"),sub:V("sc"),min:V("mn"),max:V("mx")};$("grid").innerHTML=cards()}
const VIEW={
home:()=>`<h2>Our Cake Collection</h2><div class="row"><input id="s" placeholder="Search cakes" value="${esc(fl.s)}" oninput="filt()"><select id="c" onchange="filt()">${opts([...new Set(S.cakes.map(c=>c.cat))],fl.cat,"All categories")}</select><select id="sc" onchange="filt()">${opts([...new Set(S.cakes.map(c=>c.sub))],fl.sub,"All sub categories")}</select><input id="mn" type="number" placeholder="Min Rs." value="${fl.min}" oninput="filt()"><input id="mx" type="number" placeholder="Max Rs." value="${fl.max}" oninput="filt()"></div><div class="grid" id="grid">${cards()}</div>`,
view:()=>{const c=S.cakes.find(x=>x.id==arg);return`<h2>${esc(c.name)}</h2><p>${esc(c.desc)}<br>Category: ${esc(c.cat)} | ${esc(c.sub)} | Rs.${c.price} for 500 g</p><div class="row"><div>Size<select id="sz">${opts(Object.keys(SIZE))}</select></div><div>Quantity<input id="qt" type="number" min="1" max="${c.stock}" value="1"></div></div><input id="msgc" maxlength="40" placeholder="Message on cake (Happy Birthday!)"><button ${c.stock?"":"disabled"} onclick="addCart(${c.id})">Add to Cart</button><button onclick="go('home')">Back</button>`},
login:()=>`<h2>Login</h2><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="login()">Login</button><p>Demo: kala@gmail.com / 1234, admin@sweetcrumbs.com / admin123</p>`,
reg:()=>`<h2>Register</h2><input id="n" placeholder="Full name"><input id="e" type="email" placeholder="Email"><input id="p" type="password" placeholder="Password"><button onclick="reg()">Create Account</button>`,
cart:()=>{const L=cart(),t=L.reduce((s,c)=>s+price(c),0);return`<h2>My Cart</h2><div class="tw"><table><tr><th>Cake</th><th>Size</th><th>Qty</th><th>Message</th><th>Price</th><th></th></tr>${L.map(c=>`<tr><td>${esc(S.cakes.find(k=>k.id==c.cake).name)}</td><td>${c.size}</td><td>${c.qty}</td><td>${esc(c.msg)||"-"}</td><td>Rs.${price(c)}</td><td><button class="x" onclick="S.cart=S.cart.filter(x=>x!=S.cart[${S.cart.indexOf(c)}]);save();render()">Remove</button></td></tr>`).join("")||`<tr><td colspan="6">Your cart is empty.</td></tr>`}</table></div>${L.length?`<h3>Total: Rs.${t}</h3><button onclick="go('ship')">Place Order</button>`:""}`},
ship:()=>`<h2>Shipping Address</h2><select id="ad">${me.addr.map((a,i)=>`<option value="${i}">${esc(a)}</option>`).join("")}</select><button onclick="editAddr()">Update Selected Address</button><input id="na" placeholder="Type a new address and press Add"><button onclick="addAddr()">Add New Address</button><input id="dd" type="date" min="${new Date().toISOString().slice(0,10)}"><button onclick="order()">Confirm Order (Pay on Delivery)</button>`,
my:()=>`<h2>My Orders</h2>${S.orders.filter(o=>o.uid==me.id).map(o=>`<div class="cake" style="margin:8px 0"><b>Order #${o.id}</b> | Rs.${o.total} | Delivery: ${o.date}<br>${o.items.map(i=>esc(i.name)+" ("+i.size+" x "+i.qty+(i.msg?", \""+esc(i.msg)+"\"":"")+")").join("; ")}<br>Address: ${esc(o.addr)}<br>Shipping status: <b>${o.st}</b></div>`).join("")||"<p>No orders yet.</p>"}`,
acake:()=>{const c=S.cakes.find(x=>x.id==arg)||{};return`<h2>${c.id?"Update":"Add"} Cake</h2><div class="row"><input id="an" placeholder="Name" value="${esc(c.name)}"><input id="ac" placeholder="Category" value="${esc(c.cat)}"><input id="as" placeholder="Sub category" value="${esc(c.sub)}"><input id="ap" type="number" placeholder="Price" value="${c.price||""}"><input id="ak" type="number" placeholder="Stock" value="${c.stock??""}"></div><textarea id="ad" rows="2" placeholder="Description">${esc(c.desc)}</textarea><button onclick="saveCake(${c.id||0})">${c.id?"Update":"Add"} Cake</button><div class="tw"><table><tr><th>Cake</th><th>Category</th><th>Price</th><th>Stock</th><th></th></tr>${S.cakes.map(x=>`<tr><td>${esc(x.name)}</td><td>${esc(x.cat)}</td><td>Rs.${x.price}</td><td>${x.stock}</td><td><button onclick="go('acake',${x.id})">Edit</button><button class="x" onclick="S.cakes=S.cakes.filter(y=>y.id!=${x.id});save();render()">Delete</button></td></tr>`).join("")}</table></div>`},
aord:()=>`<h2>All Orders</h2><div class="tw"><table><tr><th>#</th><th>Customer</th><th>Total</th><th>Address</th><th>Status</th></tr>${S.orders.map(o=>`<tr><td>${o.id}</td><td>${esc(S.users.find(u=>u.id==o.uid).name)}</td><td>Rs.${o.total}</td><td>${esc(o.addr)}</td><td><select onchange="setSt(${o.id},this.value)">${opts(STAGES,o.st)}</select></td></tr>`).join("")||`<tr><td colspan="5">No orders yet.</td></tr>`}</table></div>`};
function addCart(id){if(!me){go("login");return say("Please login to add cakes to your cart.",1)}const c=S.cakes.find(x=>x.id==id),q=+V("qt");if(!(q>=1)||q>c.stock)return say("Only "+c.stock+" in stock.",1);S.cart.push({uid:me.id,cake:id,size:V("sz"),qty:q,msg:V("msgc")});save();go("cart");say("Added to cart.")}
function addAddr(){if(!V("na"))return say("Type the new address first.",1);me.addr.push(V("na"));save();render();say("Address added.")}
```

```
function editAddr(){if(!V("na"))return say("Type the updated address in the box, then press Update.",1);me.addr[+V("ad")]=V("na");save();render();say("Address updated.")}
function order(){if(!me.addr.length)return say("Add a shipping address first.",1);if(!V("dd"))return say("Choose a delivery date.",1);const L=cart();
const items=L.map(c=>({name:S.cakes.find(k=>k.id==c.cake).name,size:c.size,qty:c.qty,msg:c.msg}));
L.forEach(c=>S.cakes.find(k=>k.id==c.cake).stock-=c.qty);
S.orders.push({id:S.id++,uid:me.id,items,total:L.reduce((s,c)=>s+price(c),0),addr:me.addr[+V("ad")],date:V("dd"),st:STAGES[0]}); S.cart=S.cart.filter(c=>c.uid!=me.id);save();go("my");say("Order placed. Confirmation email sent to "+me.email)}
function setSt(id,s){S.orders.find(o=>o.id==id).st=s;save();say("Order #"+id+" is now: "+s)}
function saveCake(id){const o={name:V("an"),cat:V("ac"),sub:V("as"),price:+V("ap"),stock:+V("ak"),desc:V("ad")};if(!o.name||!o.price)return say("Enter name and price.",1);if(id)Object.assign(S.cakes.find(c=>c.id==id),o);else S.cakes.push({id:S.id++,...o});save();go("acake");say("Cake saved.")}
function login(){const u=S.users.find(u=>u.email==V("e").toLowerCase()&&u.pw==$("p").value);if(!u)return say("Wrong email or password.",1);me=u;go(u.admin?"acake":"home");say("Welcome, "+u.name)}
function reg(){const e=V("e").toLowerCase();if(!V("n")||!e||$("p").value.length<4)return say("Enter name, email and a 4+ character password.",1);if(S.users.some(u=>u.email==e))return say("Email already registered.",1);const u={id:S.id++,name:V("n"),email:e,pw:$("p").value,addr:[]};S.users.push(u);save();me=u;go("home");say("Registered successfully.")}
render();
</script></body></html>
```

Output

The cake website shows cakes with filters. The user adds a customised cake to the cart, selects or updates a shipping address and places the order. The order appears in My Orders, and the administrator updates the shipping status.

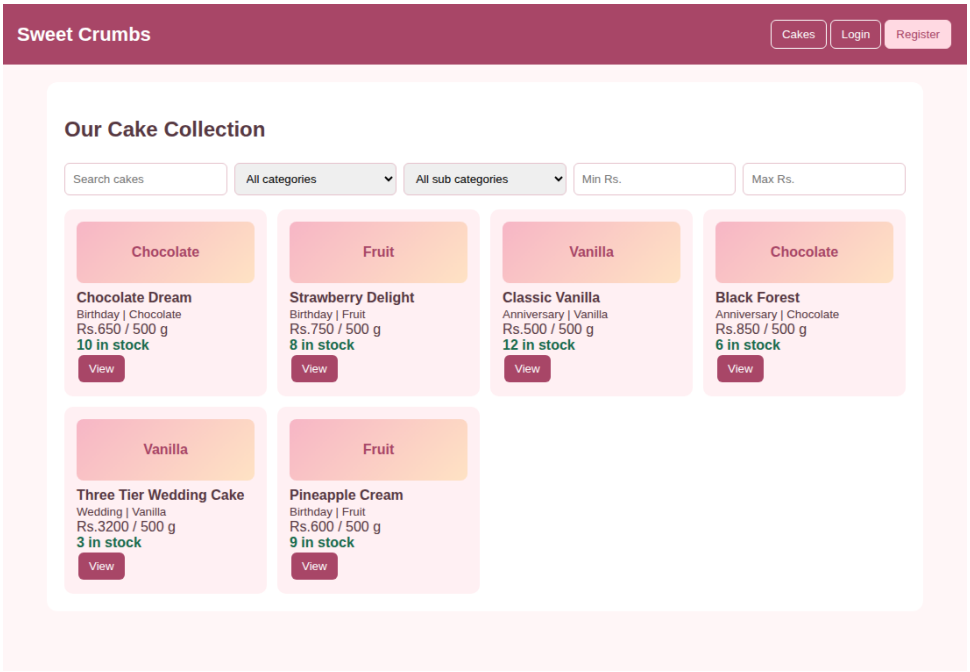


Figure 1: Cake collection with filters

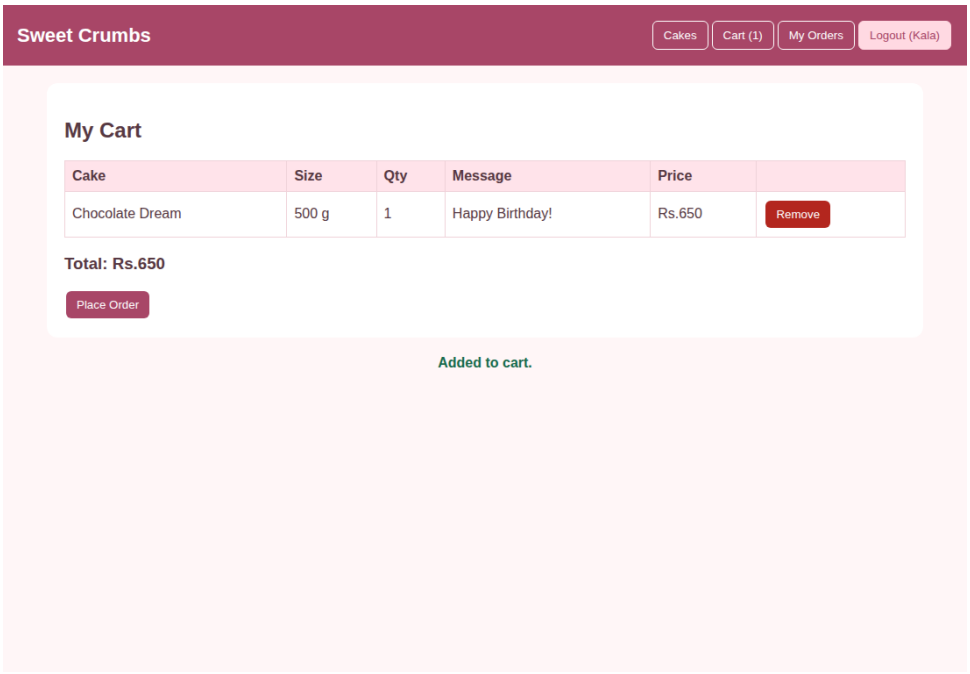


Figure 2: Cart with cake message

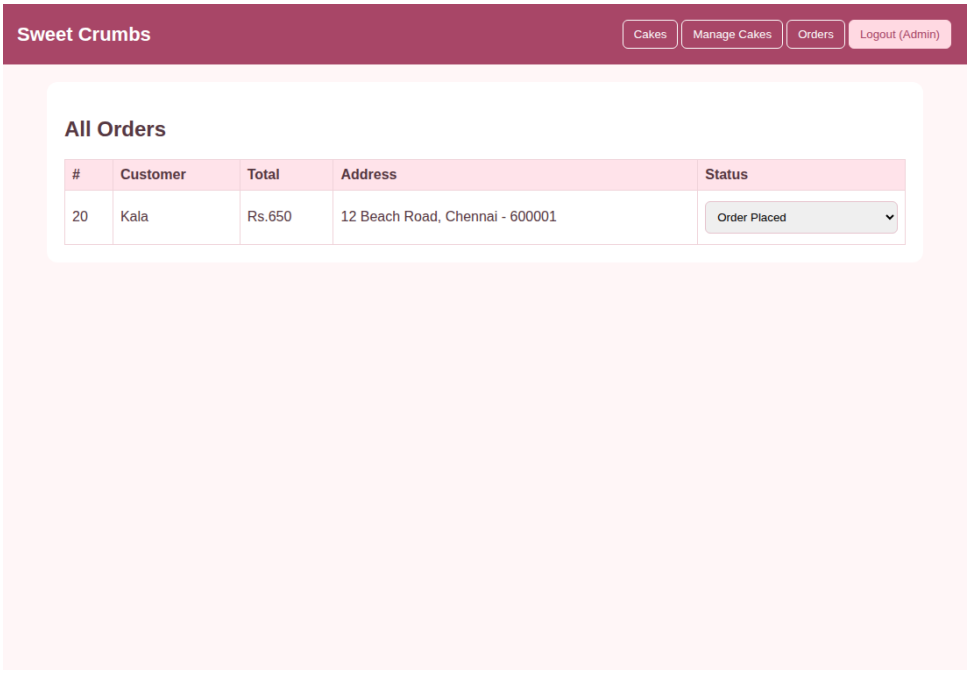


Figure 3: Admin: order status

Result

Thus, an online cake ordering website with filters, cart, shipping address and order tracking was created successfully.